

Supplementary Information

Uncertainty-aware haplotype phasing of genetic and epigenetic variation from long reads

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Supplementary Methods

These sections give the implementation-level detail of the modules that the main Methods describes only in outline, in enough detail for each algorithm to be reimplemented independently.

Supplementary Method 1. Allele-specific methylation calling (`modcall`)

Methods gives the genotyping criterion for a candidate CpG and the co-segregation criterion that a linked pair must satisfy. This section gives the parts of `modcall` that decide which candidates survive to the phasing graph.

Modification interpretation. Only the 5mC probability (modification code `m`) was used. 5hmC calls (code `h`) were ignored; under a 5mCG_5hmCG model, a cytosine called as 5hmC was counted as unmethylated whenever its 5mC probability was ≤ 0.2 . Reverse-strand calls were mapped to the forward-strand CpG coordinate, and the two strands were genotyped jointly. Non-CpG modification calls were discarded.

Linking modes and the depth bound on the neighbour scan. In *SNV-anchored* mode, used whenever an SNV VCF was supplied, a candidate CpG was accepted as a strong anchor if it satisfied the co-segregation criterion of Methods against an SNV among the next 20 variants (`-connectAdjacent`). In *methylation-only* mode, used where no informative SNV is in range, which is where methylation is most valuable for bridging, the same criterion was applied between two candidate CpGs. In both modes a pair was considered only while the number of reads spanning it exceeded $\max(6, (n_1 + n_2)/4)$, where n_1 and n_2 are the numbers of reads recorded at the two sites; a pair falling to or below that bound terminates the neighbour scan for the current site instead of merely being skipped, which prevents support from accumulating across a sparsely covered gap. Candidates never satisfying that criterion were held back as weak points and admitted only if they satisfied that criterion on at least six reads against an already accepted CpG, in an iterative expansion of two rounds (`-iterCount`, default 2).

Merging runs of CpGs. Runs of consecutive CpG coordinates are merged into a single locus represented by the first position. Because 5mC is called on one strand at a time, a CpG contributes methylated reads on the forward strand and unmethylated-looking reads on the reverse; merging consolidates both strands of a run into one locus carrying one read list, which removes the need for `phase` to reason about strand at all and lets methylation loci be voted on concurrently with SNVs by the unmodified graph algorithm. The resulting VCF records the methylated and unmethylated read names (MR, NR), the read strand and the per-site depths, and is passed to `phase` through `-mod-file`.

Supplementary Method 2. Read-level allele observations and copy-number interval detection

Variant eligibility and repeat annotation. Homozygous and filtered small-variant records were carried through to the output but did not enter the phasing graph. An optional indel-quality filter excluded indels below a user-specified QUAL threshold and marked their FILTER field; the default threshold of 0 disabled this filter. An indel was flagged as a tandem-repeat indel when the dinucleotide immediately downstream of its position occurred in five consecutive copies. Flagged indels remained in the graph and cast votes at one-tenth weight. Homozygous SVs, SVs whose breakpoint coincided with an SNV and SVs with duplicated positions were excluded. The supporting read names and length of each retained SV were recorded.

Per-class allele extraction. SNV alleles were read from the aligned base and inherited its Phred base quality. Insertion and deletion alleles were called from the CIGAR string: a read supports the ALT allele if a CIGAR insertion or deletion begins immediately after the variant position, and

the REF allele otherwise; when a read deletion spans an SNV in a homopolymer of length ≥ 3 , the SNV allele was recovered from the base following the deletion. SV alleles were assigned by scanning a window of 20 CIGAR operations (`-svWindow`) around the SV start for an insertion or deletion whose length is within 10% of the SV length (`-svThreshold`); reads listed by the SV caller as supporting the SV were also assigned the ALT allele. 5mC alleles were taken from the `modcall` read lists (methylated = allele 0, unmethylated = allele 1) restricted to reads whose strand matched the recorded strand. Soft- and hard-clipping operations longer than 5 bp were counted per reference position, separately for clips at the front (left) and back (right) end of an alignment, as evidence of alignment breakpoints for copy-number detection (below).

The copy-number-aware filter (Methods; Supplementary Fig. 1e,f) needs the boundaries of segments that recruit reads from more than one locus. Those boundaries appear in the alignments as clusters of soft- and hard-clips, and `longphase` finds them with a scanner that walks the per-position clip counts of one chromosome in coordinate order.

Input. For every reference position at which at least one alignment carries a clipping operation longer than 5 bp, two counts are recorded: u , the number of alignments clipped at their front (left) end, and d , the number clipped at their back (right) end. Positions with no clip are absent from the map. A sentinel position 30 kb beyond the last observed position is appended before the scan and removed afterwards, to close any interval left open at the end of the chromosome.

State. The scanner is in one of three states: *idle*, *step* (a copy-number transition, entered on a strong front-clip cluster) and *ramp* (the progressively nested breakpoints left by breakage–fusion–bridge cycles, entered on a weak front-clip excess). In the two active states it carries a running excess c of front- over back-clips, a candidate start, a candidate end and three adaptive thresholds derived from the front-clip count n that opened or re-anchored the interval:

$$r = n, \quad p = \begin{cases} \lfloor n/2 \rfloor, & n \geq 10 \\ 5, & n < 10 \end{cases},$$

$$f = \begin{cases} 5, & n \geq 20 \\ \lfloor n/4 \rfloor, & 10 \leq n < 20 \\ 2, & n < 10, \end{cases}$$

where r is the re-anchor threshold, p the pull-down threshold and f the decay floor.

Transitions. With constants $\Pi = 5$ (push threshold), $A = 30,000$ bp (look-ahead), $E = 30$ (extend threshold), $\Lambda = 200,000$ bp (maximum interval length) and $\Sigma = 20$ (ramp commit threshold), the scanner at position x with counts (u, d) behaves as follows:

```
idle:
  if u >= PI and c == 0:
    enter STEP; c = u - d; start = x; end = x + A
    set r, p, f from u
  elif u > d and c == 0:
    enter RAMP; c = u - d; start = x; end = x + A

step:
  if u > r:                                # stronger breakpoint: re-anchor
    start = x; end = x + A; set r, p, f from u
  c += u - d
  if c > E: end = x + A
  if d >= p:                                # pull-down reached
    emit [start, x]; reset to idle
  elif c <= f and x <= end:                 # decayed to the floor in range
    emit [start, x]; reset to idle
```

```

if x > end or c <= 0 or x - start >= LAMBDA:
    reset to idle          # candidate discarded

ramp:
if (c > SIGMA and d >= c/4) or (c <= SIGMA and d >= PI):
    emit [start, x]; reset to idle
elif u >= PI:              # promote to STEP
    c = u - d; start = x; end = x + A; set r, p, f from u
else:
    c += u - d
    if c > E: end = x + A
    if x > end or c <= 0 or x - start >= LAMBDA:
        reset to idle      # candidate discarded

```

Two behaviours are worth stating explicitly because they are not evident from the outline in the main Methods. First, a *step* interval is re-anchored, not merely extended, when a stronger front-clip cluster than the one that opened it is met: the candidate start moves to the new position and the three thresholds are recomputed, and the reported interval begins at the strongest breakpoint. Second, the decay-floor exit applies only while the scan is still inside the look-ahead window ($x \leq \text{end}$); beyond it the interval is discarded, and a candidate that decays slowly over a long distance yields nothing.

Site removal. Within each emitted interval the mismatch-rate statistic $\text{MR}(v)$ defined in Methods is computed and heterozygous variants with $\text{MR}(v) \geq 0.7$ are removed from all reads. A variant is scored only when both allele populations have a non-zero mean load; a site at which one population carries no other alternate allele is not removed on this basis.

Why MR separates the two cases. The statistic distinguishes the two situations of interest. At a genuine heterozygous site both allele populations are ordinary reads of the locus and carry comparable loads, giving $\text{MR} \approx 0.5$. At a paralogue-specific difference the alternate allele is carried by reads that originate elsewhere and therefore mismatch the reference at many other sites in the same interval as well, which drives MR towards 1.

[State how the five scanner constants were chosen (a sweep, on which data, or by inspection); none is currently justified.]

Supplementary Method 3. Calibration of the voting and read-filtering constants

The constants in this section were set empirically. Writing the ambiguity of a pair as $s = \min(P, Q) / \max(P, Q)$, which ranges from 0 for unanimous support to 1 for an even split, the voting rule has three levels: a pair votes at weight $w = 20$ when it is near-certain, that is, when $s \leq \alpha = 0.1$ with total support ≥ 1 or, irrespective of s , when one pairing has accumulated less than 1 unit of support while the other has at least 1. The pair votes at unit weight otherwise, and does not vote at all when $s > \beta = 0.7$. Each of α , β , w , the base-quality threshold $\theta = 12$ and the single-read guard parameters ($N = 3$ votes, ambiguity cut $\mu = 0.2$) was fixed by a sweep on HG002 at 10–60 $\times$ before the experiments reported here, and none was re-tuned afterwards. These sweeps were run during the development of LongPhase in 2023–2024 (base-quality threshold and β by release v1.7; α , w and the single-read guard in v2.0) and scored whole-genome phasing of earlier HG002 nanopore alignments, before the alignments evaluated here were generated and before the chromosome split used for GNN training was defined. chr17, chr21 and chr22 are therefore held out from GNN training only, not from the choice of these constants. [Argue why six scalar thresholds chosen from coarse grids do not constitute fitting; a reviewer who raises the circularity objection for the GNN will raise it here. Supplementary Fig. 9 must either be produced or removed before submission.]

Supplementary Method 4. Graph neural network: features, architecture, labels, training and export

This section expands the outline given in Methods. Supplementary Tables 1 and 2 list the node and edge features with their exact definitions, Supplementary Table 3 the architecture and inference hyperparameters, Supplementary Table 4 the training configuration, and Supplementary Figs. 4–6 illustrate window construction and the layer.

Node features. Each allele node carried 31 features in four groups (Supplementary Tables 1 and 2; architecture in Supplementary Table 3 and Supplementary Fig. 6). *Phasing evidence*: PE; the haplotype index of the allele; the haplotype indices of the reference and the alternate allele, i.e. the first and second digit of the phased genotype; whether the variant is the window trigger; whether it shares the trigger’s phase set; the allele index; the signed position offset from the trigger normalized by the largest absolute offset in the window; h_1 , h_2 , $h_1 + h_2$, the vote ratio $\max(h_1, h_2)/(h_1 + h_2)$; $\log_2(1 + h_1 + h_2)$, $\ln(1 + h_1)$, $\ln(1 + h_2)$; and $\log_2$ of the total vote weight divided by its mean over the 21 variants centred on the variant. Here h_1 and h_2 were taken from the first outgoing DOT edge of each allele when one existed and from INFO/H1, INFO/H2 otherwise. (Supplementary Table 1 lists the features by group.) *Genomic context* (computed from the reference when $-r$ is given, and zero otherwise): homopolymer length at the variant (capped at 20); GC content in ± 50 bp; the longest tandem-repeat unit count with unit size 2–4 in ± 15 bp, divided by 10 and capped at 1; Shannon entropy of the ± 15 bp sequence; the number of variants within ± 500 bp, minus one and divided by 20; and $\log_{10}(1 + d)$ for the distance d to the nearest indel, capped at $\log_{10}(10001)$. *Graph structure*: the number of distinct target positions of the variant’s outgoing DOT edges, divided by 10 and capped at 1; $\ln(1 + n_{\text{PS}})/\ln(101)$ for the number n_{PS} of variants in the variant’s phase set; $\log_{10}(1 + |\text{offset}|)$ from the trigger; the standard deviation of the variant’s outgoing DOT edge weights, divided by 100 and capped at 1; and an indicator of whether the variant is a bridge vertex, i.e. whether deleting it increases the number of connected components of the position-level, undirected window graph. *Variant class*: four one-hot indicators (SNV, indel, SV, 5mC), the SNV/indel distinction being made on $||\text{ALT}| - |\text{REF}|| > 0$.

Edge features. Edges were taken from the DOT graph, restricted to pairs both of whose endpoints lie inside the window, symmetrized by copying the features of whichever direction was present, and augmented with a self-loop per node whose features are the mean of its non-self incident edges. Each edge carried six features: the logistic-transformed vote weight $1/(1 + e^{-w})$; $\log_{10}(1 + \text{genomic distance})$; $\ln(1 + w)$; whether the two endpoints carry the same allele index; whether the two variants belong to different phase sets; and w divided by the total outgoing weight of the source allele node within the window.

Architecture. The network follows the general, powerful, scalable (GPS) graph-transformer recipe¹. Node features were standardized with a batch-normalization layer whose statistics are frozen at inference, linearly projected to a 128-dimensional embedding and passed through a GELU non-linearity. Four identical layers then each combine a local and a global branch. The local branch is a four-head GATv2 message-passing layer^{2,3} in which the attention logit of head t for edge $j \rightarrow i$ is

$$\alpha_{ij}^{(t)} \propto \exp\left(\mathbf{a}_t^\top \text{LeakyReLU}_{0.2}([\mathbf{W}_r \mathbf{x}_i + \mathbf{W}_l \mathbf{x}_j + \mathbf{W}_e \mathbf{e}_{ij}]_t)\right), \quad (1)$$

where $[\cdot]_t$ selects the 32 coordinates of head t , $\mathbf{W}_e$ is a per-layer edge encoder without bias and the normalization is a softmax over the incoming edges of i (a node with no incoming edge instead receives the uniform average of $\mathbf{W}_l \mathbf{x}_j$ over all window nodes); the head outputs are concatenated and a bias is added. The global branch is four-head scaled dot-product self-attention over all nodes of the window⁴, with no masking. The two branch outputs are added to the residual stream, followed by layer normalization, a two-layer feed-forward block (128→256→128, GELU) with a second residual connection and layer normalization. The final node embedding is concatenated

with two edge-context scalars, the sum and the non-negative maximum, over the node’s incoming edges excluding its self-loop, of the first edge feature, i.e. of the logistic-transformed vote weight $1/(1 + e^{-w})$ rather than of w itself. The pooled vector is then passed through a two-layer classifier (130→128→2) with softmax output.

The model comprises 687,358 float32 values (2,749,432 bytes), namely 124 in the input batch-normalization layer (scale, shift and the two frozen running statistics over 31 features), 4,096 in the input projection, 166,528 in each of the four GPS layers, and 17,026 in the classifier. Inference is implemented in plain C++ with dense $N \times N$ adjacency and edge tensors and no external runtime dependencies; the exact erf-based GELU must be used, because the tanh approximation changes the output by $\sim 10^{-3}$ and does not reproduce the reference implementation. Predictions for the two allele nodes of a variant were averaged. Only variants in the central zone of each window were retained: those whose genomic distance from the trigger is at most half the largest such distance in the window, $|p - p_c| \leq \frac{1}{2} \max_u |p_u - p_c|$. Because the zone is defined on genomic distance and not on variant index, the number of variants it holds depends on the local spacing: the central ± 10 variants when the window is evenly spaced, fewer when one distant variant stretches the window (the implementation writes the constant as `center_k/20` with `center_k = 10`, the same rule in the training script and in the C++ module). A variant covered by several windows received the running mean of its predictions. The same rule, with the same constant, defines which nodes carry a training loss, making the zone scored at inference identical to the zone the network was trained to score; nodes outside that zone are read as context but not scored.

Phase-set update. A variant with a mean phase-error score ≥ 0.30 was unphased while preserving its alleles: both 0|1 and 1|0 became 0/1, and the PS tag was removed. Bridge protection was disabled in the reported runs. Connectivity was then evaluated over the surviving graph edges within each original phase set. The largest connected component retained the original PS identifier; each other component received the 1-based position of its first variant as its identifier. Corrected records were written separately for small variants, SVs and methylation sites.

Training augmentations. Two label-preserving augmentations were applied to training windows only, each independently with probability 0.5: a haplotype swap, which exchanges the two allele nodes of every variant and complements the HP feature, an operation the phasing problem is invariant under; and a coverage down-scaling, which multiplies h_1 , h_2 , their sum, the outgoing-weight standard deviation and the raw edge weights by a factor drawn uniformly from $[0.4, 0.7]$ and recomputes the derived logarithmic features, applied only to windows containing at least one error node.

Labels. Labels were derived from the v5.0q benchmark restricted to chr1–chr22 small variants, the same truth set used for evaluation, and describe the phase of a variant relative to its neighbours rather than in absolute terms. Within each window the variants were grouped by phase set; in each group the first variant present in the benchmark fixed the orientation of that group and was labelled *correct*, and every later variant was labelled *correct* if its allele-to-haplotype orientation agreed with that anchor and *disagreeing* otherwise. Maximal runs of consecutive disagreeing variants were then resolved so that a run is charged to the event that created the run, not to each member. A run of length one became *unphase* if that variant’s PE was at least 0.9 and *flip* otherwise, whereas a run of length two or more had its first variant, the switch point, relabelled *break*, and all remaining members reset to *correct*. Variants absent from the benchmark, and all SV and 5mC records, for which the benchmark carries no phase truth, were left unlabelled: they are masked from the loss but remain in the graph as context, which is why a four-class window can be scored at all. The released model was trained in binary mode, in which *flip*, *unphase* and *break* are merged into a single *error* class; this matches the single action the module is allowed to take. Only labelled nodes inside the central zone of a window contribute to the loss, under the same `center_k = 10` rule used at inference.

Class balance and loss. Among the labelled central nodes of the training split, 17,812,710 (97.04%) were *correct* and 544,170 (2.96%) *error*, and 21.5% of windows contained at least one error node; the prevalence is far above the genome-wide switch-error rate because windows are opened only at high-entropy triggers and because the label is assigned relative to a window-local anchor rather than to a whole phase block. The imbalance is handled by subsampling, not by re-weighting the full distribution. Each minibatch keeps all of its error nodes and draws correct nodes uniformly at random to at most four times that number (`-ratio_corr 4`). The gradient is thus taken on an approximately 1:4 subset, and because the draw is independent at every step the network still sees the whole set of correct nodes over training. The cross-entropy on that subset is focal-weighted⁵ by $(1 - p_t)^\gamma$ with $\gamma = 1.5$ and uses label smoothing of 0.05. A differentiable surrogate for the selection metric below is implemented in the training script but was disabled for the released model ($\alpha = 0$); the reported weights were trained on the focal loss alone.

Model selection. Checkpoints were selected on the validation chromosomes by a net-gain criterion instead of validation loss or F1. At each epoch the predicted phase-error score was thresholded at each of 81 values spanning $[0.01, 0.99]$ and the score

$$S = R \left(2 - \frac{1}{P} \right) \quad (2)$$

was maximized over those thresholds subject to a recall of at least 0.005, where P and R are the precision and recall of the error class; S is positive only for $P > 0.5$ and is the net fraction of true errors removed under the accounting in which a detection is worth one error and a false detection costs one. Epochs with $P \leq 0.5$ were instead ordered by $P - 1$. The best checkpoint was epoch 44, with $S = 0.0235$ at $P = 0.626$, $R = 0.058$ and a threshold of 0.868; no later epoch improved on that checkpoint before early stopping. On the held-out chromosomes chr17, chr21 and chr22 that checkpoint attained an accuracy of 0.941 with error-class precision, recall and F1 of 0.258, 0.499 and 0.340 under the argmax decision, and 0.946 with 0.273, 0.468 and 0.345 at the F1-optimal threshold of 0.537 tuned on the validation chromosomes, over 41,378 labelled error nodes. The score S governed checkpoint selection; the deployment threshold was chosen separately using the end-to-end phasing trade-off described below.

Deployment-threshold selection. Phase-error score thresholds B from 0.05 to 0.95 in steps of 0.05 were evaluated on 60 HG002 nanopore runs, comprising ten down-sampling replicates at each of 10, 12, 14, 16, 18 and 20 $\times$. Each threshold was scored, relative to phasing without GNN correction, by the fraction of switch errors removed and by the cost of correction, the number of phased SNVs withheld divided by the number of switch errors removed. Because the two trade off against each other, the criterion was the largest fraction of switch errors removed at a cost of at most ten withheld SNVs per removed error. For the released model, $B = 0.30$ removed 33.1% of switch errors at a cost of 8.3 withheld SNVs per removed error (60 runs pooled, whole genome). On the held-out chromosomes chr17, chr21 and chr22, the same threshold removed 21.2% of switch errors at a cost of 10.5, slightly above the limit; the threshold was not re-selected on these chromosomes (Supplementary Table 15). This criterion concerns the final phasing output and is distinct from the classifier-based criterion used to rank training checkpoints.

Export and numerical agreement. The released weights are a single training checkpoint. A dense re-implementation replaces the sparse graph operators with $N \times N$ adjacency and edge tensors, transfers the trained tensors (splitting the fused attention projection into separate Q, K and V matrices), and exports to ONNX at opset 17 with N as a dynamic axis. The logistic transform of the vote weight and the self-loop features are folded into the input convention at this point, and the exported graph and the compiled implementation both receive edge features with those already applied. The ONNX initializers are then read, the architecture dimensions recovered from the tensor shapes, and both written into a generated C++ header as a base64-encoded little-endian float32 blob in a fixed tensor order; the compiled forward pass reads the blob in that order and refuses

to run if its length disagrees with the expected parameter count. Agreement was checked at each link of the chain: the dense rewrite against the PyTorch Geometric model, and the ONNX graph against the dense rewrite, on a real training window, with an acceptance bound of 10^{-3} on the absolute difference in predicted phase-error score; and the compiled C++ against ONNX Runtime on six synthetic graphs of 3–120 nodes at edge densities 0.1–0.4, including one graph with a single edge and no other incoming edges, where the largest absolute difference was 3×10^{-7} against an acceptance bound of 10^{-5} .

Supplementary Method 5. Read haplotagging

`longphase haplotag` assigns each alignment to a haplotype using the phased VCFs produced by phasing with or without GNN correction. The main Methods gives the decision rule; this section gives the evidence model, which differs from the one used inside `phase` in ways that matter to anyone comparing haplotagged fractions between tools.

Alignment selection. Primary alignments with mapping quality ≥ 1 (`-q`, `-qualityThreshold`) are considered. Supplementary alignments are tagged only under `-tagSupplementary`; secondary alignments never are. Any pre-existing HP, PS and PQ tags are removed from every record before tagging, which makes re-tagging an already-tagged file safe. Unmapped records are passed through unchanged.

Evidence collection. Each alignment is traversed once along its CIGAR, and four classes of evidence are collected into two per-haplotype tallies (n_1, n_2) :

- *SNVs* are read from `M`, `=` and `X` operations; a read carrying the allele phased to haplotype k adds 1 to n_k .
- *Small indels* are scored by allele length rather than by sequence comparison. Writing ℓ_1 and ℓ_2 for the lengths of the two phased alleles at the site, a read that carries an indel there supports the haplotype whose allele has $\ell \neq 1$, and a read that does not supports the haplotype whose allele has $\ell = 1$. When both or neither allele has length 1 the site casts no vote.
- *SVs* are re-derived from the alignment, not taken from the caller’s read list: a window of $\pm$ -`svWindow` CIGAR operations (default 20) around the current operation is scanned for an insertion or deletion whose length is within `-svThreshold` (default 0.10) of the SV length, and the first match adds 1 to the haplotype the SV is phased to.
- *Allele-specific 5mC* sites are matched by read name, not by position: the methylated-read list `INFO/MR` of the phased 5mC VCF adds 1 to the haplotype that record is phased to, for each read it names. The unmethylated list `NR` is not consulted; a 5mC site contributes evidence only to the reads that are methylated at it.

Every informative observation contributes exactly 1, whatever its class. This is deliberately different from the read-based correction inside `phase`, where SNV and SV alleles contribute 1, indel and tandem-repeat-indel alleles 0.1 and 5mC alleles nothing (Methods). A read can therefore be tagged in `haplotag` on indel and methylation evidence that would have been discounted during phasing.

Decision and phasing quality. Let $m = \max(n_1, n_2)$ and $\mu = \min(n_1, n_2)$. The read is assigned to the majority haplotype when $m/(m + \mu) \geq$ `-p`, `-percentageThreshold` (default 0.6) and is otherwise left untagged. A Phred-scaled phasing quality is computed for every alignment as

$$\text{PQ} = \begin{cases} 0 & m = 0, \\ 40 & \mu = 0, \\ \left\lfloor -10 \log_{10} \frac{\mu}{m + \mu} \right\rfloor & \text{otherwise,} \end{cases} \quad (3)$$

so a read whose alleles are unanimous receives the ceiling value 40 and a read split 2:1 receives 4. PQ is written as an integer auxiliary tag PQ:i alongside HP:i and PS:i, and only on tagged reads; untagged reads carry none of the three. PQ is not reported in the main text and no analysis in this paper uses it.

Phase-set veto. The phase sets of all informative variants on the read are counted. If more than one distinct PS is represented, the read is left untagged regardless of how one-sided its haplotype tally is, because a majority computed across two independent phase sets has no meaning. This veto bounds the taggable fraction from above and is the main reason a read spanning a block boundary is not tagged. The PS value written on a tagged read is that of the first phased variant at or after the alignment start, whether or not that variant is informative on the read; because the veto has already required every informative variant to share one phase set, the two coincide except when that first variant is uninformative on the read and belongs to a different block, a boundary case.

Supplementary Method 6. Evaluation metrics

Metrics were computed with `longphase compare` (Supplementary Fig. 8), a C++ re-implementation of the `whatshap compare` algorithm⁶.

Block intersection. For each chromosome, heterozygous records common to the truth and a query are intersected by position and by REF/ALT alleles. The intersection is partitioned into maximal runs that lie in one truth phase set and one query phase set simultaneously; runs of length one carry no phase information and are discarded. The switch errors and the Hamming distance of a query are computed inside its surviving blocks but with different denominators: the switch error rate divides SNV switch errors by the number of consecutive SNV pairs assessed, whereas the Hamming distance divides misassigned variants, of every class present in both files, by the number of compared variants; the phased fraction is counted before this partition, as the number of truth heterozygous variants whose position, REF and first ALT match a query record that is heterozygous and phased, over all truth heterozygous variants of the class.

Switch errors and flips. Each block yields two binary strings, the truth haplotype assignment p_0 and the query assignment q_0 of its variants. Both are converted to *switch encoding*, the string of relative changes between consecutive positions, which removes the arbitrary orientation of the block. Switch errors are the Hamming distance between the two switch-encoded strings. The decomposition into isolated switches and flips follows WhatsHap’s `compute_switch_flips`: a maximal run of k consecutive disagreements in switch space is charged as $\lfloor k/2 \rfloor$ flips and $k \bmod 2$ switches. A single variant assigned to the wrong haplotype thus counts as one flip, not two switches. The *switch error rate* is the total number of switch errors divided by the number of assessed consecutive pairs.

Hamming distance. For each block the query strings (q_0, q_1) are compared with the truth strings (t_0, t_1) in both orientations, and the smaller of $H(t_0, q_0) + H(t_1, q_1)$ and $H(t_0, q_1) + H(t_1, q_0)$ is halved to give the number of misassigned variants in that block. Block values are summed and expressed as a percentage of compared variants.

Block N50. Block spans in bp are sorted in decreasing order and the N50 is the span at which the cumulative sum first reaches half the total, which is the span-weighted median. It is computed twice: once per chromosome, and once over the block spans of all chromosomes pooled. The values plotted in Figs. 2c,g,k, 3b,c and 5c are the pooled, genome-wide N50 of the query, which does not depend on the truth set.

Supplementary Notes

Supplementary Note 1. Reading the phasing graph export

With `-dot`, `longphase phase` writes one file per chromosome, `<prefix>.<chrom>.dot`, containing one line per voting pair, `P.a -> Q.b [label=w]`, where `P` and `Q` are 1-based positions, `a`, `b` $\in \{1, 2\}$ index the reference (1) and alternate (2) allele, and `w` is the weighted haplotype vote accumulated at `P` for that allele at the time the vote was cast. The companion VCF carries, for every phased variant, the fields `INFO/H1`, `INFO/H2` (weighted votes for haplotype 1 and 2) and `INFO/PE` (their normalized Shannon entropy). These exports record the graph evidence used to construct the local GNN windows; the GNN stage of `phase` receives the same edges and weights in memory, so `-dot` is needed only to inspect them.

Supplementary Note 2. Reproducing the demonstration

The repository ships a 500-kb demonstration (`chr1:121.5–122.0 Mb`; `demo/demo.sh`) that performs methylation calling, SNV-only and four-class phasing with and without GNN correction, and comparison against a truth VCF, and prints switch errors, phased SNV counts and Hamming distance for the four configurations. Total runtime is under two minutes on four threads.

Supplementary Tables

Supplementary Table 1: Node features of the graph neural network. Each allele node of a window carries 31 features; # is the 1-based index in the feature vector, listed here by group, not by index. h_1 , h_2 : weighted haplotype votes (taken from the DOT edge labels when available, otherwise from INFO/H1, INFO/H2); PE: phasing entropy; PS: phase set; “centre” is the trigger variant of the window. Genomic-context features are computed from the reference when -r is given and are zero otherwise.

#	Feature	Definition
<i>Phasing evidence (16)</i>		
1	PE	normalized Shannon entropy of (h_1, h_2) , 0–1 bit
2	HP	haplotype of this allele node (genotype index)
3	gt_ref	haplotype index of the reference allele: first digit of the phased GT (0 or 1)
4	gt_alt	haplotype index of the alternate allele: second digit of the phased GT (0 or 1)
5	is_center	1 if the variant is the window trigger
6	is_same_ps	1 if the variant shares the trigger’s PS
7	allele	0 for the reference-allele node, 1 for the alternate-allele node
8	rel_offset	$(\text{pos} - \text{pos}_c) / \max \text{pos} - \text{pos}_c $ over the window, signed
9	H1	h_1
10	H2	h_2
11	total_votes	$h_1 + h_2$
12	vote_ratio	$\max(h_1, h_2) / (h_1 + h_2)$
13	log_total	$\log_2(1 + h_1 + h_2)$
14	log_h1	$\ln(1 + h_1)$
15	log_h2	$\ln(1 + h_2)$
25	rel_vote_depth	$\log_2((h_1 + h_2) / \overline{(h_1 + h_2)}_{\pm 10 \text{ variants}})$
<i>Genomic context (6)</i>		
16	hp_len	homopolymer run length at the variant (capped at 20)
17	gc_content	GC fraction in ± 50 bp
18	str_context	longest tandem-repeat unit count (unit 2–4 bp) in ± 15 bp, /10, capped at 1
19	seq_entropy	Shannon entropy of the ± 15 bp sequence (bits)
20	neigh_dens	$(\text{number of variants within } \pm 500 \text{ bp} - 1) / 20$
24	dist_near_indel	$\log_{10}(1 + \text{distance to nearest indel})$, capped at $\log_{10}(10001)$
<i>Graph structure (5)</i>		
21	out_degree	number of distinct target positions of outgoing DOT edges, /10, capped at 1
22	ps_block_size	$\ln(1 + \text{variants in the PS}) / \ln(101)$
23	log_center_dist	$\log_{10}(1 + \text{pos} - \text{pos}_c)$
26	out_weight_std	s.d. of outgoing DOT edge weights, /100, capped at 1
27	is_bridge	1 if removing the variant disconnects the window graph
<i>Variant class (4, one-hot)</i>		
28–31	is_SNV, is_indel, is_SV, is_5mC	variant class indicators

Supplementary Table 2: Edge features of the graph neural network. Edges are taken from the DOT export, restricted to node pairs inside the window, symmetrized, and complemented by one self-loop per node whose features are the mean over the node’s incident edges. w is the DOT edge label (accumulated vote weight).

#	Feature	Definition
1	weight	$1/(1 + e^{-w})$
2	log_dist	$\log_{10}(1 + \text{pos}_{\text{dst}} - \text{pos}_{\text{src}})$
3	log_weight	$\ln(1 + w)$
4	is_same_hp	1 if the two allele nodes have the same allele index
5	is_inter_ps	1 if the two variants belong to different phase sets
6	weight_ratio	$w / (\text{sum of outgoing edge weights of the source allele node})$

Supplementary Table 3: Architecture and inference hyperparameters of the graph neural network.

Component	Setting
Input normalization	BatchNorm over 31 node features (frozen statistics at inference)
Input projection	Linear $31 \rightarrow 128$, GELU
Layers	4 GPS layers, hidden size 128
Local branch	GATv2, 4 heads $\times$ 32, LeakyReLU slope 0.2, per-layer edge encoder $6 \rightarrow 128$
Global branch	multi-head self-attention, 4 heads, Q/K/V/O projections $128 \rightarrow 128$
Fusion	$x \leftarrow x + \text{local} + \text{global}$; LayerNorm ($\varepsilon = 10^{-5}$)
Feed-forward	$128 \rightarrow 256 \rightarrow 128$, GELU, residual, LayerNorm
Classifier	concat $[h_i, \sum_j \sigma(w_{ij}), \max_j \sigma(w_{ij})]$ (130) $\rightarrow$ 128 (GELU) $\rightarrow$ 2, softmax; σ logistic, self-loop excluded
Parameters	687,358 (float32, 2.75 MB, embedded in the binary)
Window	trigger PE ≥ 0.8 ; ± 20 variants (≤ 41 variants, ≤ 82 allele nodes; a 256-node cap exists but is never reached at this window size)
Decision	unphase if mean $P(\text{error}) \geq 0.30$ over the two allele nodes and all covering windows
Post-processing	split disconnected phase sets (largest component keeps PS); bridge protection off
Training	see Supplementary Table 4

Supplementary Table 4: Training configuration of the graph neural network. Values are those recorded in the released checkpoint and its training log; every setting is the training script’s default except `-binary`, which was passed explicitly. “Scored nodes” are labelled allele nodes inside the central zone of a window, the only nodes that contribute to the loss.

Item	Setting
<i>Data</i>	
Source runs	HG002 ONT R10.4.1, four-class co-phasing, 10/12/14/16/18/20 × × 10 down-sampling replicates (60 runs)
Window extraction	<code>-pe_threshold 0.8, -window 20</code> , GRCh38 no-alt analysis set
Truth set	v5.0q benchmark, chr1–chr22 small variants
Split (by chromosome)	train chr1–chr14, chr18–chr20; validation chr15, chr16; test chr17, chr21, chr22
Windows / scored nodes	train 602,788 / 18,356,880; val 67,880 / 1,509,576; test 51,448 / 1,367,580
Class prevalence (train)	correct 17,812,710 (97.04%), error 544,170 (2.96%); 21.5% of windows contain an error node
Labels	anchor-relative within each PS; single disagreement → <i>unphase</i> if $PE \geq 0.9$ else <i>flip</i> ; run of ≥ 2 → first variant <i>break</i> , rest <i>correct</i> ; merged to one <i>error</i> class
Masked from loss	variants absent from the benchmark; all SV and 5mC records (the truth set holds small variants only); nodes outside the central zone
<i>Optimization</i>	
Loss	focal cross-entropy, $\gamma = 1.5$, label smoothing 0.05, on a per-batch subset of all error nodes plus $\leq 4\times$ as many randomly drawn correct nodes
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\varepsilon = 10^{-8}$, weight decay 10^{-3}
Learning rate	3×10^{-4} , no warm-up, cosine annealing to 10^{-5} over 100 epochs
Batch size	64 windows, batched as disjoint graphs
Regularization	dropout 0.4 (local branch, global branch, FFN; inactive at inference); gradient-norm clipping 1.0
Graph preprocessing	reverse edge added per DOT edge with identical features (undirected)
Augmentation (train only)	haplotype swap, $p = 0.5$; coverage down-scaling $\times U[0.4, 0.7]$, $p = 0.5$, error-bearing windows only
Epochs	budget 100, early stopping after 25 epochs without improvement; stopped at 69
Seed	42 (Python and PyTorch generators)
Environment	PyTorch 2.11.0, PyTorch Geometric 2.7.0, Python 3.12, CUDA 13.0, cuDNN 9.19, one NVIDIA GeForce RTX 5090
<i>Selection and export</i>	
Selection criterion	max over 81 thresholds in $[0.01, 0.99]$ of $R(2 - 1/P)$ subject to $R \geq 0.005$, on chr15–chr16; $P - 1$ where $P \leq 0.5$
Selected checkpoint	epoch 44; $S = 0.0235$ at $P = 0.626$, $R = 0.058$, threshold 0.868
Held-out test (chr17, 21, 22)	argmax: accuracy 0.941, error-class $P/R/F_1 = 0.258/0.499/0.340$; tuned threshold: 0.946, 0.273/0.468/0.345; $n = 41,378$ error nodes
Deployment threshold	0.30; phased-SNV/switch-error trade-off (Supplementary Method 4)
Export	ONNX opset 17, dynamic N ; base64 float32 blob compiled into the binary; C++ vs. ONNX Runtime max. absolute difference 3×10^{-7}

Supplementary Table 6: Selected published tools, pipelines and studies that use LongPhase.
Compiled from Europe PMC (80 indexed citing articles) and OpenAlex (96 citing works) for
10.1093/bioinformatics/btac058, September 2026; the “use” column quotes or paraphrases the citing
article’s Methods.

Work	Journal, year	Use of LongPhase
<i>Variant callers and pipelines</i>		
ClairS ¹⁰	Nat. Methods 2026	default germline phasing of tumour and normal alignments (<code>-whatshap_for_phasing</code> the alternative); haplotagging selectable
ClairS-TO ¹¹	Nat. Commun. 2025	default intermediate phasing and haplotagging of the tumour alignment (WhatsHap selectable)
Clair3-RNA ¹²	Nat. Commun. 2025	phasing in the long-read RNA pipeline
Clair3 (accelerated) ¹³	Bioinformatics 2026	selectable intermediate phasing and haplotagging (<code>-longphase_for_phasing</code> ; WhatsHap the default)
DeepVariant local haplotagging ¹⁴	Nat. Commun. 2024	not used; cited as one of the haplotaggers (with WhatsHap and Margin) that Clair3 and Medaka employ
Severus ¹⁵	Nat. Biotechnol. 2025	one of the documented routes to the haplotagged BAMs it consumes (WhatsHap the other)
EPI2ME wf-human-variation ¹⁶	ONT workflow	intermediate per-contig phasing (<code>phase_contig</code>); WhatsHap for final phasing and haplotagging
HiFi-somatic-WDL ¹⁷	PacBio workflow	optional phasing tool
cuteHap ¹⁸	Adv. Sci. 2026	read phasing before haplotype-aware SV calling
SpecImmune ¹⁹	Adv. Sci. 2026	joint SNV–SV phasing of immune genes
<i>Biological and clinical studies</i>		
Fluctuating DNA methylation in cancer ²⁰	Nature 2025	read phasing (v1.7) and <code>modcall</code> for CpG methylation
Advanced cancer cohort (POG) ²¹	Cell Genomics 2024	discussed as phasing method for tumours
Pediatric epilepsy episignatures ²²	Nat. Commun. 2024	phasing of Clair3 and SV calls
GNAS transcripts in PHP1B ²³	PNAS 2025	phasing of Clair3 calls
3-hour genome for neonatal risk ²⁴	Genet. Med. Open 2024	real-time phasing during sequencing
Ultrarapid diagnostics in critical care ²⁵	Eur. J. Hum. Genet. 2025	read phasing (v1.0.7)
HYDIN in primary ciliary dyskinesia ²⁶	Eur. J. Hum. Genet. 2024	phasing of Clair3 calls
Antisense oligonucleotides in APBD ²⁷	Nucleic Acids Res. 2025	phasing of Clair3 calls
Native CpG methylation fine-mapping ²⁸	HGG Adv. 2026	phasing and haplotagging (chosen for speed and SV support)
Canadian national platform benchmark ²⁹	Front. Genet. 2026	phasing inside the Clair3 workflow
Pallas’s cat genome ³⁰	NAR Genom. Bioinform. 2023	haplotagging for allele-specific methylation

Work	Journal, year	Use of LongPhase
<i>Methods evaluated against LongPhase</i>		
HiPhase ³¹	Bioinformatics 2024	comparison method
Ralphi ³²	Genome Res. 2025	comparison method (fastest runtime on ONT/PacBio)
GCphase ³³	BMC Bioinformatics 2024	comparison method
Duet ³⁴	BMC Bioinformatics 2022	comparison method for SV phasing

Supplementary Table 5: Software and data used in the benchmarks.

Item	Version	Notes
HG002 nanopore R10.4.1 reads	ONT giab_lsk114_2022.12	FLO-PRO114M, SQK-LSK114, 4 kHz; Dorado sup@v4.0.0 with 5mCG_5hmCG@v2
Reference	GRCh38	GCA_000001405.15_GRCh38_no_alt_analysis_set; no-alt analysis set, no decoys
minimap2	2.24-r1122	-y -ax map-ont (alignment by Oxford Nanopore; reads with mean quality ≥ 10)
PEPPER-Margin-DeepVariant ⁷	r0.8	SNV and indel calls (unphased genotypes); call_variant with -ont_r10_q20 (nanopore) or -hifi (HiFi), one call set per coverage and replicate
Sniffles2	2.8.0	-output-rnames
GIAB v4.2.1 HG002	v4.2.1	GRCh38 small-variant benchmark with phase transferred from a hifiasm assembly; HG002_GRCh38_1_22_v4.2.1_benchmark_hifiasm_v11_phasetransfer.vcf.gz, biallelic phased heterozygous records
GIAB v5.0q HG002 benchmark ^{8,9}	v5.0q	GRCh38 small-variant benchmark derived from the T2T-HG002 v1.1 assembly; HG002_GRCh38_v5.0q_smvar.vcf.gz, chr1-chr22 records
WhatsHap	2.8	phase -ignore-read-groups -reference; -only-snvs for SNV comparisons; default indel phasing for indel comparisons
HapCUT2	1.3.4	extractHAIRS -ont 1, then HAPCUT2 with default parameters (default pruning); input restricted to heterozygous SNVs
Margin	2.3.1	SNV-only comparison, 10–20 $\times$; phasing parameters allParams.phase_vcf.ont.json; input restricted to SNVs
Ralphi	v0.0.1 (commit d7d17b1)	SNV-only comparison, 10–20 $\times$; model ralphi.v1.long.pt, ONT configuration with max_coverage 15, mapq 20, read selection and bad-read filtering
MethPhaser	0.0.4	run on the uncorrected SNV-only LongPhase 2 output (same SNV phasing as the joint SNV+5mC run), replicate 1 [command line]
HG002 PacBio HiFi reads	Revio, PacBio release 2023-10-31	ccs 7.0.0, lima 2.7.1, primrose 1.4.0; aligned with pbmm2 1.10.0 (48 $\times$); down-sampled with samtools 1.23.1 (seed 1) to 10–40 $\times$, 48 $\times$ used as 50 $\times$; PEPPER-Margin-DeepVariant r0.8 -hifi, called separately at every coverage
LongPhase 1.0	v1.0	-ont; commit fe8c1fdff11ba2a082d363a7ffcb5c923cacc316
LongPhase 2	v2.1, commit 46ba470	Nanopore mode (-pb for HiFi); GNN correction with defaults

Supplementary Table 7: Input call sets per coverage. Heterozygous (het.) and homozygous (hom.) calls entering the phasing runs; nanopore values at 10–20× are means (s.d.) over ten down-sampling replicates. Small variants from PEPPER-Margin-DeepVariant, SVs from Sniffles2 2.8.0, allele-specific 5mC sites from longphase modcall. SVs and 5mC were not called on the HiFi data. Homozygous SNVs include the PEPPER candidates genotyped homozygous reference (refCall).

Platform	Cov.	Het. SNV	Hom. SNV	Het. indel	Het. SV	Hom. SV	Het. 5mC
Nanopore	10×	2,340,598 (3,826)	3,265,243 (8,954)	283,612 (381)	14,866 (130)	7,684 (57)	283,932 (1,026)
Nanopore	12×	2,506,797 (2,922)	3,227,137 (8,307)	315,147 (511)	16,119 (69)	7,874 (33)	361,073 (2,274)
Nanopore	14×	2,586,505 (3,184)	3,173,111 (5,212)	332,537 (343)	16,707 (67)	8,008 (31)	408,058 (1,847)
Nanopore	16×	2,621,584 (2,784)	3,115,855 (8,105)	342,202 (505)	17,089 (68)	8,121 (17)	431,895 (976)
Nanopore	18×	2,636,024 (1,840)	3,051,563 (5,197)	347,703 (324)	17,326 (80)	8,201 (34)	444,081 (1,158)
Nanopore	20×	2,640,440 (2,408)	3,004,806 (5,806)	350,682 (328)	17,487 (43)	8,247 (32)	452,032 (1,079)
Nanopore	30×	2,636,000	2,902,347	356,240	17,800	8,473	474,512
Nanopore	40×	2,617,161	2,902,110	358,059	17,764	8,635	485,070
Nanopore	50×	2,598,083	2,914,706	360,142	17,602	8,754	493,774
Nanopore	60×	2,579,823	2,929,398	361,119	17,474	8,834	499,496
HiFi	10×	2,367,609	2,081,641	425,786	–	–	–
HiFi	20×	2,485,185	2,318,132	442,122	–	–	–
HiFi	30×	2,468,170	2,428,526	432,578	–	–	–
HiFi	40×	2,438,078	2,508,667	425,166	–	–	–
HiFi	50×	2,410,815	2,566,571	420,096	–	–	–

Supplementary Table 8: SNV-only phasing of HG002 nanopore R10.4.1 data by six configurations. Scored against the v5.0q VCF (chr1–22, no benchmark BED applied). Values at 10–20× are means (s.d.) over ten down-sampling replicates; 30–60×, one replicate. Phased SNVs are given as a percentage of the 2,398,880 heterozygous SNVs of the benchmark. Margin and Ralphie were run at 10–20× only; GCphase was not included (Methods). Ralphie’s phase blocks sum to 4.3–5.8 Gb, more than the length of the autosomes, so its block N50 is not comparable with that of the other tools.

Tool	Cov.	Phased SNV (%)	Switch errors	Switch error rate (%)	Hamming (%)	Blocks	N50 (Mb)
LongPhase 2	10×	77.83 (0.07)	1,518 (29)	0.082 (0.002)	5.33 (0.25)	7,143 (30)	0.84 (0.01)
	12×	84.48 (0.06)	1,335 (54)	0.066 (0.003)	5.41 (0.38)	5,917 (52)	1.10 (0.02)
	14×	87.87 (0.05)	1,137 (39)	0.054 (0.002)	4.88 (0.58)	5,284 (55)	1.29 (0.03)
	16×	89.58 (0.03)	1,011 (27)	0.047 (0.001)	4.20 (0.26)	4,875 (44)	1.44 (0.03)
	18×	90.50 (0.04)	918 (18)	0.042 (0.001)	3.64 (0.26)	4,608 (58)	1.53 (0.03)
	20×	90.94 (0.02)	823 (43)	0.038 (0.002)	3.21 (0.34)	4,406 (46)	1.61 (0.02)
	30×	91.50	680	0.031	1.82	3,793	1.92
	40×	91.54	602	0.027	1.71	3,411	2.18
	50×	91.46	588	0.027	1.48	3,198	2.51
	60×	91.35	496	0.023	1.53	3,018	2.86
LongPhase 2 without GNN	10×	78.10 (0.07)	2,259 (39)	0.121 (0.002)	7.04 (0.29)	6,056 (46)	0.92 (0.01)
	12×	84.71 (0.06)	1,995 (60)	0.098 (0.003)	7.21 (0.42)	4,878 (39)	1.21 (0.02)
	14×	88.07 (0.05)	1,706 (50)	0.081 (0.002)	6.43 (0.55)	4,314 (52)	1.40 (0.02)
	16×	89.75 (0.03)	1,514 (40)	0.070 (0.002)	5.51 (0.21)	3,964 (43)	1.54 (0.03)
	18×	90.65 (0.03)	1,374 (36)	0.063 (0.002)	4.75 (0.22)	3,740 (34)	1.62 (0.03)
	20×	91.08 (0.02)	1,235 (45)	0.057 (0.002)	4.12 (0.38)	3,564 (35)	1.70 (0.02)
	30×	91.61	1,004	0.046	2.40	3,007	1.97
	40×	91.64	878	0.040	2.37	2,624	2.29
	50×	91.55	883	0.040	1.86	2,389	2.57
	60×	91.43	743	0.034	1.75	2,225	2.91
WhatsHap (SNVs only)	10×	78.74 (0.07)	4,383 (65)	0.233 (0.003)	7.60 (0.39)	5,960 (49)	0.95 (0.01)
	12×	85.35 (0.05)	4,229 (126)	0.207 (0.006)	7.99 (0.40)	4,897 (45)	1.24 (0.03)
	14×	88.71 (0.05)	3,848 (127)	0.181 (0.006)	7.01 (0.42)	4,435 (47)	1.41 (0.02)
	16×	90.37 (0.04)	3,561 (134)	0.165 (0.006)	6.37 (0.33)	4,169 (52)	1.53 (0.02)
	18×	91.27 (0.03)	3,291 (134)	0.151 (0.006)	5.23 (0.39)	4,009 (34)	1.60 (0.04)
	20×	91.68 (0.02)	3,067 (147)	0.140 (0.007)	4.40 (0.35)	3,904 (38)	1.65 (0.02)
	30×	92.26	2,524	0.114	2.56	3,499	1.88
	40×	92.36	2,139	0.097	3.05	3,208	2.15
	50×	92.37	2,010	0.091	2.40	3,034	2.47
	60×	92.34	1,785	0.081	2.75	2,912	2.72
HapCUT2	10×	78.78 (0.09)	5,018 (108)	0.266 (0.006)	7.77 (0.39)	5,878 (55)	0.95 (0.01)
	12×	85.41 (0.05)	4,948 (150)	0.242 (0.007)	8.06 (0.32)	4,837 (41)	1.25 (0.02)
	14×	88.76 (0.07)	4,652 (104)	0.219 (0.005)	7.10 (0.52)	4,382 (48)	1.44 (0.03)
	16×	90.44 (0.04)	4,344 (96)	0.201 (0.004)	6.13 (0.41)	4,122 (50)	1.54 (0.02)
	18×	91.32 (0.05)	4,068 (121)	0.186 (0.006)	5.32 (0.29)	3,969 (39)	1.61 (0.04)

Tool	Cov.	Phased SNV (%)	Switch errors	Switch error rate (%)	Hamming (%)	Blocks	N50 (Mb)
	20×	91.70 (0.03)	3,825 (140)	0.174 (0.006)	4.27 (0.35)	3,872 (42)	1.66 (0.02)
	30×	92.27	3,280	0.148	2.61	3,482	1.89
	40×	92.37	2,916	0.132	2.90	3,203	2.16
	50×	92.38	2,568	0.116	2.20	3,034	2.46
	60×	92.34	2,309	0.104	2.11	2,911	2.72
Margin	10×	78.53 (0.07)	1,286 (31)	0.069 (0.002)	0.47 (0.04)	18,177 (100)	0.26 (0.00)
	12×	85.13 (0.06)	1,347 (32)	0.066 (0.002)	0.62 (0.10)	13,654 (88)	0.40 (0.01)
	14×	88.47 (0.05)	1,331 (46)	0.063 (0.002)	0.72 (0.05)	11,181 (69)	0.52 (0.01)
	16×	90.14 (0.03)	1,302 (49)	0.061 (0.002)	0.79 (0.11)	9,738 (58)	0.62 (0.01)
	18×	91.04 (0.03)	1,281 (41)	0.059 (0.002)	0.87 (0.10)	8,818 (49)	0.71 (0.00)
	20×	91.47 (0.02)	1,254 (60)	0.057 (0.003)	0.93 (0.11)	8,129 (46)	0.78 (0.01)
Ralph	10×	78.52 (0.08)	3,784 (104)	0.201 (0.006)	7.48 (0.27)	5,692 (49)	5.31 (7.10)
	12×	85.14 (0.06)	3,694 (164)	0.181 (0.008)	8.32 (0.53)	4,536 (34)	7.97 (8.42)
	14×	88.50 (0.06)	3,452 (102)	0.163 (0.005)	7.71 (0.63)	4,000 (58)	25.31 (16.84)
	16×	90.17 (0.04)	3,176 (104)	0.147 (0.005)	7.60 (0.52)	3,681 (36)	30.67 (12.45)
	18×	91.07 (0.03)	2,979 (163)	0.137 (0.007)	7.14 (0.51)	3,463 (30)	39.14 (14.83)
	20×	91.50 (0.02)	2,874 (96)	0.131 (0.004)	6.56 (0.53)	3,314 (37)	47.11 (6.72)

Supplementary Table 9: The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark. Switch errors and Hamming distance on SNVs; all LongPhase 2 runs with GNN correction, phasing SNVs alone or with indels (+indel) or indels and SVs (+indel+SV). Values at 10–20× are means (s.d.) over ten replicates. Phased indels as a percentage of benchmark heterozygous indels (v5.0q).

Configuration	Cov.	Switch errors v4.2.1	Switch errors v5.0q	Hamming v4.2.1 (%)	Hamming v5.0q (%)	Phased indels (%)
LongPhase 2, SNVs	10×	1,820 (26)	1,518 (29)	5.30 (0.24)	5.33 (0.25)	–
	20×	1,656 (31)	823 (43)	3.12 (0.36)	3.21 (0.34)	–
	30×	1,635	680	1.80	1.82	–
	40×	1,663	602	1.64	1.71	–
	50×	1,669	588	1.43	1.48	–
	60×	1,676	496	1.54	1.53	–
LongPhase 2, +indels	10×	1,972 (24)	1,704 (32)	6.80 (0.44)	6.85 (0.44)	30.42 (0.05)
	20×	1,807 (30)	996 (35)	5.75 (0.42)	5.78 (0.40)	41.03 (0.03)
	30×	1,777	822	4.72	4.86	43.38
	40×	1,758	724	4.22	4.37	44.39
	50×	1,752	664	3.13	3.26	45.02
	60×	1,765	591	3.53	3.46	45.35
LongPhase 2, +indels+SVs	10×	1,977 (24)	1,711 (30)	6.81 (0.41)	6.85 (0.41)	30.42 (0.05)
	20×	1,806 (33)	1,001 (33)	6.04 (0.51)	6.06 (0.50)	41.02 (0.03)
	30×	1,778	838	4.98	5.07	43.38
	40×	1,767	739	4.20	4.38	44.39
	50×	1,760	673	3.29	3.39	45.01
	60×	1,772	592	3.50	3.46	45.34
WhatsHap, SNVs only	10×	2,773 (47)	4,383 (65)	7.44 (0.37)	7.60 (0.39)	–
	20×	1,972 (29)	3,067 (147)	4.21 (0.38)	4.40 (0.35)	–
	30×	1,850	2,524	2.45	2.56	–
	40×	1,810	2,139	2.92	3.05	–
	50×	1,811	2,010	2.23	2.40	–
	60×	1,801	1,785	2.61	2.75	–
WhatsHap, SNVs + indels	10×	3,538 (48)	5,174 (78)	12.00 (0.34)	12.16 (0.36)	32.86 (0.03)
	20×	2,580 (32)	3,683 (122)	11.34 (0.59)	11.45 (0.57)	44.31 (0.04)
	30×	2,345	3,027	9.67	9.71	47.14
	40×	2,221	2,572	10.43	10.40	48.52
	50×	2,171	2,346	8.32	8.46	49.36
	60×	2,156	2,139	9.20	9.29	49.84
HapCUT2	10×	3,050 (33)	5,018 (108)	7.59 (0.38)	7.77 (0.39)	–
	20×	2,346 (40)	3,825 (140)	4.09 (0.35)	4.27 (0.35)	–
	30×	2,202	3,280	2.46	2.61	–
	40×	2,171	2,916	2.72	2.90	–
	50×	2,133	2,568	2.03	2.20	–
	60×	2,088	2,309	2.02	2.11	–

Supplementary Table 10: Co-phasing configurations of LongPhase 2, without and with GNN correction. SNV metrics against v5.0q; phased indels as a percentage of benchmark heterozygous indels. Means (s.d.) over ten replicates at 10 and 20×; one replicate at 30 and 60×. Every coverage is in Supplementary Data 1.

Classes	GNN	Cov.	Phased indel (%)	Switch errors	Switch error rate (%)	Hamming (%)	N50 (Mb)
SNV	no	10×	–	2,259 (39)	0.121 (0.002)	7.04 (0.29)	0.92 (0.01)
		20×	–	1,235 (45)	0.057 (0.002)	4.12 (0.38)	1.70 (0.02)
		30×	–	1,004	0.046	2.40	1.97
		60×	–	743	0.034	1.75	2.91
	yes	10×	–	1,518 (29)	0.082 (0.002)	5.33 (0.25)	0.84 (0.01)
		20×	–	823 (43)	0.038 (0.002)	3.21 (0.34)	1.61 (0.02)
		30×	–	680	0.031	1.82	1.92
		60×	–	496	0.023	1.53	2.86
+SV	no	10×	–	2,292 (47)	0.123 (0.002)	7.23 (0.24)	0.93 (0.01)
		20×	–	1,262 (44)	0.058 (0.002)	4.44 (0.35)	1.72 (0.01)
		30×	–	1,031	0.047	2.64	1.98
		60×	–	745	0.034	2.09	2.93
	yes	10×	–	1,541 (34)	0.083 (0.002)	5.45 (0.24)	0.85 (0.01)
		20×	–	842 (44)	0.039 (0.002)	3.52 (0.32)	1.63 (0.02)
		30×	–	706	0.032	1.96	1.93
		60×	–	498	0.023	1.68	2.90
+5mC	no	10×	–	2,247 (45)	0.120 (0.002)	7.13 (0.33)	0.94 (0.01)
		20×	–	1,228 (41)	0.056 (0.002)	4.38 (0.52)	1.84 (0.03)
		30×	–	1,015	0.046	2.53	2.12
		60×	–	736	0.034	1.82	3.14
	yes	10×	–	1,541 (32)	0.083 (0.002)	5.46 (0.27)	0.86 (0.01)
		20×	–	824 (40)	0.038 (0.002)	3.38 (0.35)	1.75 (0.02)
		30×	–	692	0.032	2.01	2.07
		60×	–	497	0.023	1.60	3.09
+indel	no	10×	30.79 (0.04)	2,913 (52)	0.156 (0.003)	10.87 (0.39)	1.18 (0.02)
		20×	41.25 (0.03)	1,646 (47)	0.075 (0.002)	9.50 (0.38)	2.45 (0.05)
		30×	43.55	1,316	0.060	8.02	2.96
		60×	45.48	953	0.043	6.38	4.35
	yes	10×	30.42 (0.05)	1,704 (32)	0.092 (0.002)	6.85 (0.44)	0.97 (0.02)
		20×	41.03 (0.03)	996 (35)	0.046 (0.002)	5.78 (0.40)	2.01 (0.06)
		30×	43.38	822	0.037	4.86	2.44
		60×	45.35	591	0.027	3.46	3.69
+indel+SV	no	10×	30.78 (0.04)	2,934 (51)	0.157 (0.003)	11.00 (0.39)	1.19 (0.02)
		20×	41.24 (0.03)	1,657 (51)	0.076 (0.002)	9.76 (0.43)	2.48 (0.04)
		30×	43.54	1,323	0.060	8.45	2.99
		60×	45.48	952	0.043	6.51	4.37
	yes	10×	30.42 (0.05)	1,711 (30)	0.092 (0.002)	6.85 (0.41)	0.98 (0.02)
		20×	41.02 (0.03)	1,001 (33)	0.046 (0.002)	6.06 (0.50)	2.04 (0.05)
		30×	43.38	838	0.038	5.07	2.42
		60×	45.34	592	0.027	3.46	3.70
+indel+5mC	no	10×	30.79 (0.05)	2,877 (48)	0.154 (0.003)	10.98 (0.37)	1.21 (0.02)
		20×	41.26 (0.03)	1,596 (46)	0.073 (0.002)	9.70 (0.55)	2.70 (0.08)
		30×	43.55	1,317	0.060	8.01	3.19
		60×	45.49	937	0.043	7.33	5.03
	yes	10×	30.48 (0.05)	1,724 (27)	0.093 (0.001)	7.04 (0.45)	0.99 (0.02)
		20×	41.08 (0.03)	972 (31)	0.045 (0.001)	5.99 (0.60)	2.21 (0.06)
		30×	43.41	842	0.038	5.27	2.69
		60×	45.39	593	0.027	3.60	4.10
all four	no	10×	30.79 (0.05)	2,897 (49)	0.155 (0.003)	11.08 (0.43)	1.21 (0.02)
		20×	41.26 (0.03)	1,610 (52)	0.074 (0.002)	9.88 (0.62)	2.74 (0.07)
		30×	43.54	1,316	0.060	8.33	3.19
		60×	45.48	932	0.043	7.23	5.10
	yes	10×	30.47 (0.05)	1,726 (30)	0.093 (0.002)	7.06 (0.45)	1.00 (0.02)
		20×	41.08 (0.03)	981 (34)	0.045 (0.002)	6.13 (0.57)	2.24 (0.06)
		30×	43.41	863	0.039	5.50	2.71
		60×	45.38	585	0.027	3.34	4.10

Supplementary Table 11: SNV-only phasing of HG002 PacBio HiFi (Revio) data. One replicate per coverage, scored against v5.0q. The network was trained on nanopore data only and applied without retraining. Variants were called separately at every coverage (Methods).

Tool	Cov.	Phased SNV (%)	Switch errors	Switch error rate (%)	Hamming (%)	Blocks	N50 (kb)
LongPhase 2	10×	87.02	1,070	0.052	0.78	15,916	341
	20×	91.73	945	0.043	1.15	12,190	489
	30×	91.92	885	0.040	0.82	11,164	541
	40×	91.89	869	0.040	0.78	10,557	573
	50×	91.58	740	0.034	0.89	10,215	584
LongPhase 2 without GNN	10×	87.12	1,328	0.064	0.96	15,708	346
	20×	91.84	1,268	0.058	1.28	11,980	492
	30×	92.02	1,137	0.052	1.02	10,926	545
	40×	91.98	1,153	0.052	0.98	10,347	580
	50×	91.66	988	0.045	1.10	10,010	588
WhatsHap (SNVs only)	10×	87.43	2,655	0.128	1.03	15,804	340
	20×	92.20	2,504	0.114	1.21	12,090	478
	30×	92.40	2,331	0.106	0.92	11,082	525
	40×	92.37	1,954	0.089	0.87	10,512	561
	50×	92.05	1,687	0.077	0.68	10,186	577

Supplementary Table 13: Location of SNV switch errors by truth set and benchmark region. **a**, switch errors by truth set and region; **b**, by genomic stratum within the v5.0q benchmark regions; **c**, switch errors shared by the three tools. SNV-only phasing of HG002 nanopore R10.4.1 data; LongPhase 2 with GNN correction. A variant pair is assigned to a region when both of its variants lie in it, so region counts do not sum to the total. *Shared*, inside both sets of benchmark regions; *v5.0q only*, inside the v5.0q regions only; *outside*, outside both. Means over ten replicates at 10 and 20×; one replicate at 30 and 60×. Rates are switch errors per assessed SNV pair. Per-run values for every coverage, region and stratum are in Supplementary Data 1.

Tool	Cov.	v4.2.1		v5.0q					
		All	Shared	All	Inside v5.0q	Rate inside v5.0q (%)	Shared	v5.0q only	Outside (rate, %)
LongPhase 2	10×	1,820	1,445	1,518	938	0.0527	805	41	438 (1.19)
	20×	1,656	1,169	823	271	0.0130	196	30	446 (0.95)
	30×	1,635	1,127	680	117	0.0056	62	25	454 (0.99)
	60×	1,676	1,178	496	66	0.0032	33	15	348 (0.81)
WhatsHap	10×	2,773	2,235	4,383	1,992	0.1112	1,483	266	2,030 (4.63)
	20×	1,972	1,372	3,067	723	0.0347	340	240	2,065 (3.75)
	30×	1,850	1,238	2,524	444	0.0211	134	205	1,829 (3.34)
	60×	1,801	1,245	1,785	235	0.0112	63	88	1,337 (2.61)
HapCUT2	10×	3,050	2,432	5,018	2,445	0.1364	1,686	322	2,175 (4.93)
	20×	2,346	1,633	3,825	1,264	0.0606	613	315	2,262 (4.10)
	30×	2,202	1,492	3,280	937	0.0446	393	268	2,053 (3.74)
	60×	2,088	1,435	2,309	676	0.0321	273	148	1,430 (2.78)

b Within the v5.0q benchmark regions, by GIAB v3.6 stratum

Tool	Cov.	Segmental duplication	Low mappability or seg. dup.	Tandem repeat	Homopolymer	Satellite	Not difficult
LongPhase 2	10×	51	55	5	1	2	604
	20×	37	39	3	0	2	135
	30×	29	29	0	0	0	45
	60×	18	19	0	0	0	20
WhatsHap	10×	264	279	33	6	12	1,079
	20×	232	239	29	5	10	240
	30×	202	205	16	5	3	91
	60×	80	82	15	0	0	38
HapCUT2	10×	292	308	54	19	12	1,040
	20×	275	283	45	21	10	244
	30×	247	254	36	19	6	100
	60×	120	124	25	21	3	40

c Switch errors at identical SNV pairs across tools, replicate 1

Truth	Cov.	LongPhase 2	WhatsHap	HapCUT2	All three	All three (% of LongPhase 2)
v4.2.1	10×	1,829	2,775	3,041	1,145	63
	20×	1,680	1,922	2,292	1,405	84
	30×	1,635	1,850	2,202	1,474	90
	60×	1,676	1,801	2,088	1,556	93
v5.0q	10×	1,510	4,258	4,920	543	36
	20×	922	2,735	3,552	289	31
	30×	680	2,524	3,280	236	35
	60×	496	1,785	2,309	183	37

Supplementary Table 14: Phase accuracy of indels against v5.0q. HG002 nanopore R10.4.1 data, co-phasing runs. Indel-pair switch errors are counted over consecutive variant pairs of an intersected block that include at least one indel; the indel Hamming distance uses the block orientation chosen on all variants. Truth VCF used in full; the last column restricts the indel-pair rate to the v5.0q benchmark regions. Means (s.d.) over ten replicates at 10 and 20×; one replicate at 30 and 60×. Per-run values are in Supplementary Data 1.

Configuration	Cov.	Phased indels	Indel-pair switch errors	Indel-pair rate (%)	SNV+indel rate (%)	Indel Hamming (%)	Indel-pair rate, inside v5.0q (%)
LongPhase 2, +indel	10×	157,148 (247)	300 (23)	0.101 (0.008)	0.090 (0.001)	6.78 (0.46)	0.080 (0.007)
	20×	211,933 (164)	206 (10)	0.052 (0.003)	0.045 (0.001)	5.72 (0.37)	0.032 (0.003)
	30×	224,081	179	0.043	0.037	4.89	0.023
	60×	234,254	130	0.030	0.027	3.52	0.015
same, no GNN	10×	159,017 (231)	495 (31)	0.165 (0.010)	0.150 (0.003)	10.83 (0.41)	0.133 (0.009)
	20×	213,062 (161)	305 (10)	0.076 (0.002)	0.073 (0.002)	9.49 (0.34)	0.050 (0.003)
	30×	224,926	236	0.056	0.058	8.08	0.031
	60×	234,940	183	0.042	0.042	6.40	0.022
LongPhase 2, four classes	10×	157,408 (252)	311 (21)	0.104 (0.007)	0.091 (0.001)	6.98 (0.43)	0.084 (0.005)
	20×	212,187 (161)	200 (11)	0.050 (0.003)	0.044 (0.001)	6.08 (0.53)	0.030 (0.003)
	30×	224,236	181	0.043	0.039	5.59	0.022
	60×	234,414	138	0.031	0.027	3.46	0.015
WhatsHap, +indel	10×	169,742 (173)	2,685 (57)	0.840 (0.018)	0.350 (0.005)	12.54 (0.32)	0.734 (0.014)
	20×	228,884 (187)	3,379 (55)	0.788 (0.013)	0.272 (0.006)	11.91 (0.62)	0.700 (0.013)
	30×	243,513	3,764	0.827	0.262	10.37	0.751
	60×	257,426	4,377	0.911	0.253	9.81	0.858

Supplementary Table 15: SNV-only phasing on the chromosomes held out from GNN training (chr17, chr21 and chr22). HG002 nanopore R10.4.1 data; the phased VCFs of the genome-wide analysis scored with `longphase compare -regions=chr17,chr21,chr22` against each truth VCF in full. Switch errors of LongPhase 2 as mean (s.d.); switch error rates in %. *Removed*, switch errors removed by GNN correction relative to LongPhase 2 without it; *cost*, phased SNVs withheld per removed switch error (the deployment criterion requires at most 10; Supplementary Method 4). Means over ten replicates at 10–20×; one replicate at 30–60×. Margin and Ralph were run at 10–20× only.

Truth	Cov.	LongPhase 2		Switch error rate (%)					GNN correction	
		Switch errors	Rate (%)	without GNN	WhatsHap	HapCUT2	Margin	Ralph	Removed (%)	Cost
v5.0q	10×	126 (16)	0.127	0.163	0.360	0.416	0.099	0.347	22.0	10.4
	12×	116 (16)	0.107	0.133	0.339	0.390	0.095	0.322	20.2	11.5
	14×	92 (12)	0.082	0.106	0.288	0.334	0.089	0.282	23.1	10.6
	16×	90 (10)	0.078	0.099	0.258	0.316	0.083	0.248	21.3	10.0
	18×	84 (8)	0.072	0.089	0.243	0.303	0.081	0.249	19.5	10.1
	20×	71 (16)	0.060	0.076	0.233	0.297	0.076	0.243	20.9	9.7
	30×	83	0.070	0.081	0.196	0.212	–	–	13.5	12.2
	40×	80	0.067	0.080	0.179	0.187	–	–	15.8	7.3
	50×	61	0.052	0.061	0.152	0.182	–	–	15.3	11.9
	60×	37	0.031	0.039	0.144	0.154	–	–	19.6	15.1
v4.2.1	10×	134 (8)	0.149	0.178	0.215	0.225	0.080	0.195	16.4	9.3
	12×	128 (8)	0.131	0.153	0.180	0.193	0.081	0.172	14.4	9.1
	14×	126 (6)	0.123	0.138	0.166	0.176	0.085	0.155	11.0	9.9
	16×	121 (7)	0.116	0.128	0.146	0.159	0.086	0.142	9.6	9.3
	18×	121 (5)	0.115	0.122	0.138	0.151	0.092	0.137	5.8	12.4
	20×	120 (6)	0.113	0.118	0.134	0.147	0.093	0.132	4.2	12.3
	30×	118	0.110	0.114	0.128	0.140	–	–	3.3	9.2

Truth	Cov.	LongPhase 2		Switch error rate (%)					GNN correction	
		Switch errors	Rate (%)	without GNN	WhatsHap	HapCUT2	Margin	Ralphi	Removed (%)	Cost
	40×	112	0.104	0.105	0.118	0.128	–	–	0.9	61.0
	50×	117	0.109	0.109	0.117	0.125	–	–	0.0	–
	60×	117	0.109	0.110	0.118	0.126	–	–	0.8	24.0

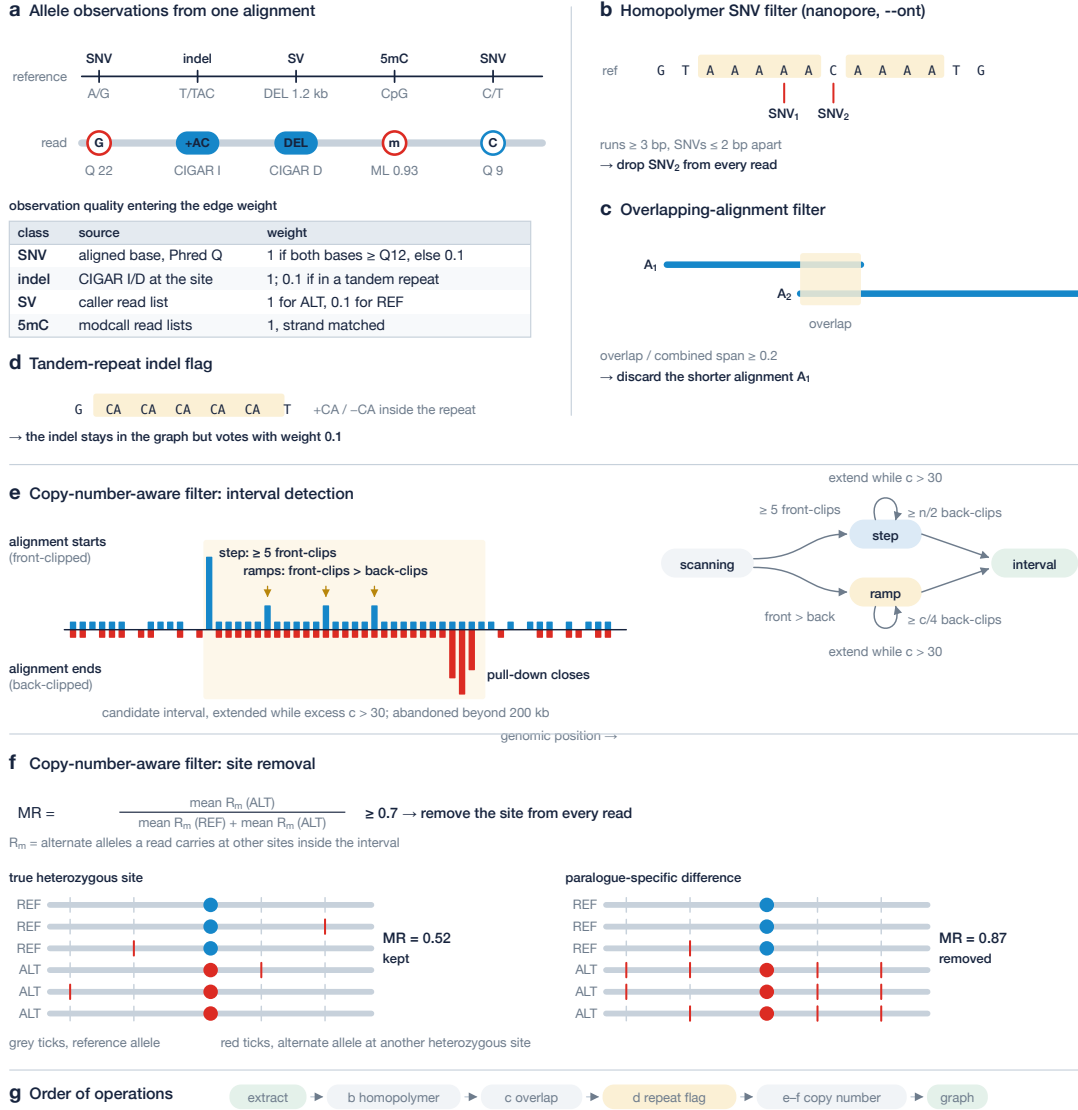
Supplementary Data 1. `supplementary.xlsx`: per-run metrics behind Figs. 2–6, Supplementary Figs. 10–16 and Supplementary Tables 7–11, 13 and 14, with one row per coverage and down-sampling replicate. Sheet `Variant_Calling`, input call sets; `SNV_Detail`, SNV-only phasing by every tool on nanopore and HiFi data; `co-phase_Detail`, the six co-phasing configurations of LongPhase 2 without (`longphase_v2.0.1`) and with (`longphase_v2.1`) GNN correction, and WhatsHap with indels; `SNV_switch_by_region`, SNV switch errors and assessed pairs of SNV-only runs per truth set, benchmark region and GIAB v3.6 stratum; `SNV_switch_coincidence`, switch errors at identical SNV pairs across tools (replicate 1); `Indel_accuracy`, indel and SNV switch errors, assessed pairs and Hamming distance of the co-phasing runs, without and within the v5.0q benchmark regions. Blank cells are runs that were not performed.

Supplementary Table 12: SNVs unphased by GNN correction, classified by the T2T-HG002 v1.1 assembly. Nanopore 60×, replicate 1, SNV-only phasing. “Aligned” means inside the dipcall BED in which both assembled haplotypes align 1:1 to GRCh38 (GRCh38_HG2-T2TQ100-V1.1_dipcall-z2k.dip.bed); variant status from the matching dipcall VCF. The background is a random sample of 50,000 SNVs that stayed phased. Top, overall classes; bottom, share outside the 1:1 alignment per chromosome. v5.0q is derived from the same assembly, so this analysis extends coverage rather than providing an independent truth set.

Class	SNVs	Share (%)
not aligned 1:1 (unassessed)	49,475	88.23
aligned, assembly carries no variant	3,421	6.10
aligned, assembly confirms a variant	3,180	5.67
total	56,076	100.00
aligned 1:1, unphased by GNN correction	6,601 / 56,076	11.77
aligned 1:1, phased before GNN correction (sample)	46,318 / 50,000	92.64
inside the v5.0q benchmark BED	549	—
of which benchmark variant	413	75.23

Chromosome	Phased before GNN correction <i>n</i>	not aligned (%)	Unphased by GNN correction <i>n</i>	not aligned (%)	Ratio
chr1	3,787	7.05	3,877	88.26	12.5
chr2	3,713	3.42	1,930	92.69	27.1
chr3	3,467	4.99	2,410	80.12	16.1
chr4	3,355	3.37	1,188	75.93	22.5
chr5	3,184	2.73	1,249	61.89	22.7
chr6	3,135	2.90	676	88.61	30.6
chr7	2,949	5.49	3,060	77.65	14.1
chr8	2,707	5.13	1,872	91.19	17.8
chr9	2,342	12.34	1,689	92.30	7.5
chr10	2,478	4.16	1,530	89.80	21.6
chr11	2,639	13.19	10,385	99.06	7.5
chr12	2,420	9.13	4,374	99.73	10.9
chr13	1,801	6.11	2,268	96.69	15.8
chr14	1,590	2.01	339	52.21	26.0
chr15	1,482	8.70	1,096	78.01	9.0
chr16	1,496	8.29	2,660	85.64	10.3
chr17	1,512	11.11	1,373	67.81	6.1
chr18	1,679	24.84	6,988	98.10	3.9
chr19	1,130	0.09	138	42.75	475.0
chr20	1,219	11.16	1,627	88.57	7.9
chr21	781	20.23	545	88.99	4.4
chr22	756	14.68	321	88.79	6.0
chrX	263	44.11	4,006	68.50	1.6
chrY	115	52.17	475	19.16	0.4

Supplementary Figures

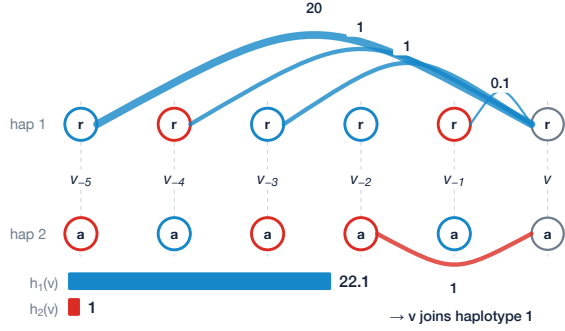


Supplementary Fig. 1: Read-level allele observations and pre-graph filters. **a**, For every alignment, each overlapped variant yields an allele observation with an associated quality: SNVs from the aligned base (Phred quality), indels from CIGAR operations starting at the site, SVs from CIGAR operations of matching length or from the caller's read list, and 5mC from the `modcall` read lists. The table summarizes how each class enters the edge weights. **b**, Homopolymer SNV filter, applied under `--ont`: when two SNVs both lie in homopolymers of length ≥ 3 and are ≤ 2 bp apart, the downstream SNV is removed. **c**, Overlapping-alignment filter for supplementary alignments of one read. **d**, Indels in tandem repeats are flagged and vote with weight 0.1. **e**, Copy-number-aware filter, interval detection. Per-position counts of alignments starting (front-clipped, above the axis) and ending (back-clipped, below) are scanned by a three-state detector (idle, step, ramp; right): a *step* boundary (≥ 5 front-clips at one position) marks a copy-number transition; a *ramp* boundary (front-clips exceeding back-clips without reaching that threshold) marks the nested breakpoints left by breakage–fusion–bridge cycles. Both states carry a running excess c of front- over back-clips and extend the candidate end by a 30 kb look-ahead while $c > 30$; a step closes at an adaptive pull-down threshold ($n/2$ for an opening front-clip count $n \geq 10$, else 5) or when c decays to its floor, a ramp closes when back-clips reach $c/4$. Candidates longer than 200 kb are abandoned. **f**, Copy-number-aware filter, site removal. Inside an interval, each read is scored by its mismatch load R_m , the number of heterozygous sites in the interval at which it carries the alternate allele, and each heterozygous site by $\text{MR} = \bar{R}_m^{\text{ALT}} / (\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}})$. At a genuine heterozygous site both allele populations are ordinary reads of the locus and $\text{MR} \approx 0.5$; at a paralogue-specific difference the alternate allele is carried by reads that originate elsewhere and mismatch the reference at many other sites in the same interval, driving $\text{MR} \rightarrow 1$. Sites with $\text{MR} \geq 0.7$ are removed from all reads; sites at which either mean is zero are never removed. **g**, Order in which the filters are applied; the copy-number-aware filter (**e,f**) acts last, immediately before graph construction.

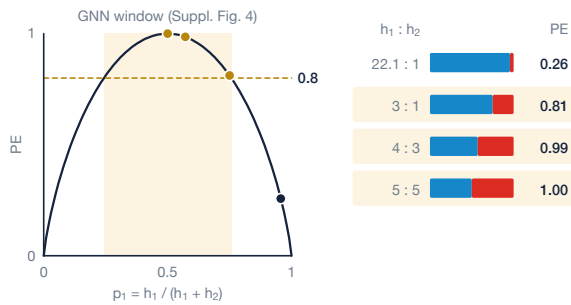
a Allele-pair support between u and v



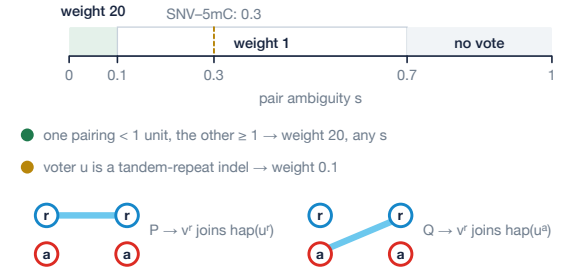
c Weighted votes from up to $k = 35$ upstream variants



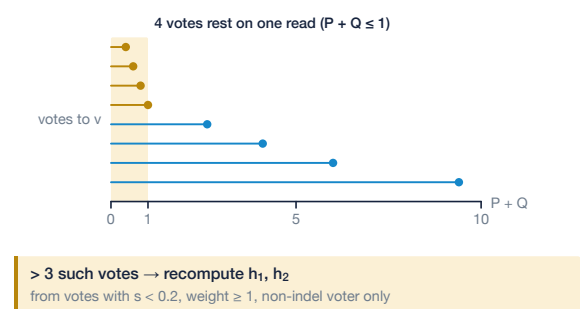
e Phasing entropy $PE = -\sum p_i \log_2 p_i$



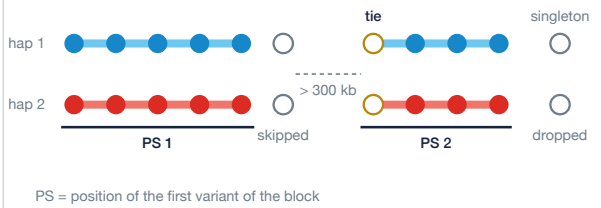
b Vote rule for the pair (u , v)



d Single-read guard

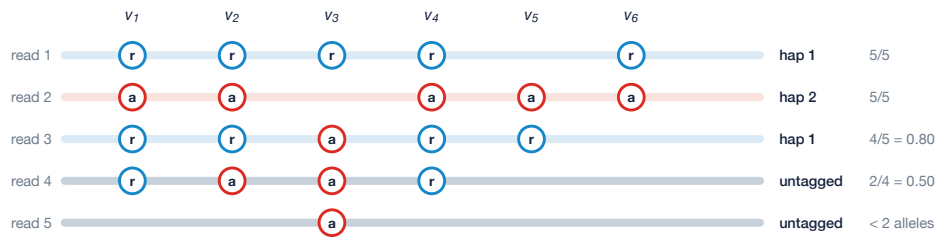


f Block formation



Supplementary Fig. 2: Pairwise allele support, weighted voting and phasing entropy. **a**, Between two variants u and v the reads accumulate four weights ($w_{rr}, w_{ra}, w_{ar}, w_{aa}$), each read adding 1 when both bases have quality $\geq Q12$ and 0.1 otherwise. $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$ are the supports for the two possible pairings and $s = \min(P, Q) / \max(P, Q)$ their ambiguity. **b**, The pair votes for the better-supported pairing: at weight 20 when $s \leq 0.1$ with $P + Q \geq 1$ or, irrespective of s , when one pairing has less than 1 unit of support and the other at least 1; at weight 1 otherwise; and not at all when $s > 0.7$ (0.3 for SNV-5mC pairs, which also cast no vote when $P + Q < 1$). A voting variant that is a tandem-repeat indel votes at weight 0.1. The vote names a haplotype for v relative to the haplotype already assigned to u . **c**, Variant v receives weighted votes from up to $k = 35$ upstream variants that share a read with it, at any distance; line thickness encodes vote weight. The weighted totals h_1 and h_2 decide the haplotype; a tie opens a new phase block. **d**, Guard against a single erroneous ultra-long read: among the votes received by v , those resting on a single read ($P + Q \leq 1$, amber) are counted, and if more than three exist, h_1 and h_2 are recomputed from the votes with $s < 0.2$, weight ≥ 1 and a non-indel voter only. **e**, Phasing entropy of the vote totals, $PE = -\sum_i p_i \log_2 p_i$ with $p_i = h_i / (h_1 + h_2)$, as a function of the vote split (left) and for four examples (right); values ≥ 0.8 (shaded) open a window for the neural network (Supplementary Fig. 4). PE, h_1 and h_2 are written to the VCF as INFO/PE, INFO/H1 and INFO/H2. **f**, Block formation. Variants are visited in coordinate order; a variant whose next heterozygous variant lies more than 300 kb away (**-distance**) is skipped and casts no votes, a variant that receives no informative vote or a tie (amber) starts a new phase set when it lies beyond the last position reached by an accepted edge and is otherwise left unphased, and blocks holding a single variant are dropped (grey). Haplotype labels propagate along the block as the phased genotypes 0|1 and 1|0; the phase set (PS) is the position of its first variant.

a Step 1: read assignment to haplotypes

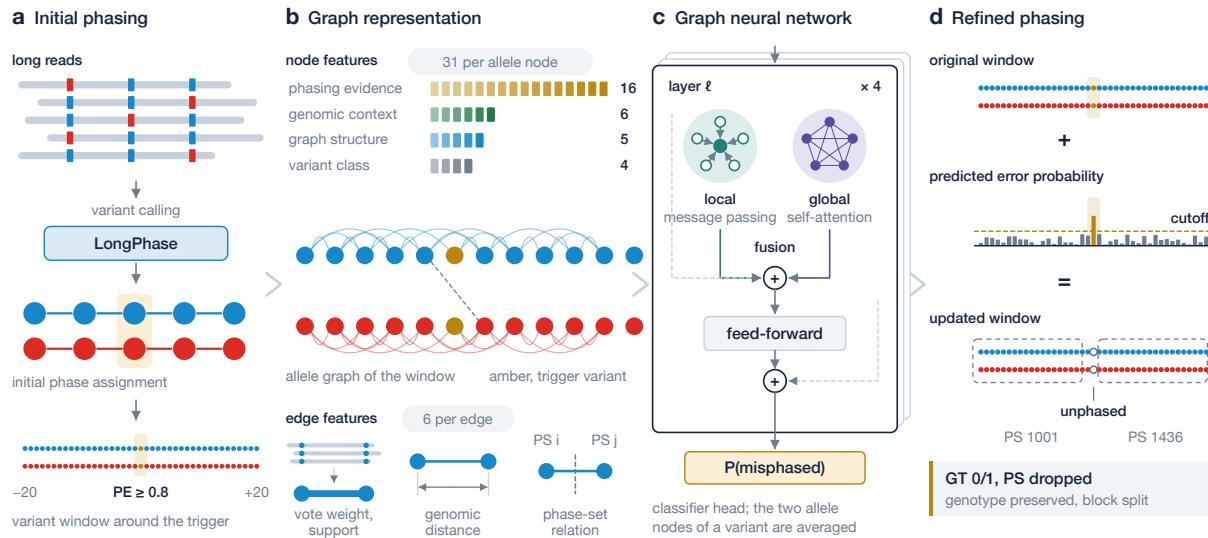


tagged if $\max(n_1, n_2) / (n_1 + n_2) > 0.65$ and weighted evidence > 1
 allele weights: SNV 1, SV 1, indel 0.1, 5mC 0 (so ≥ 2 SNV/SV alleles are needed)

b Step 2: phase re-derivation from tagged reads

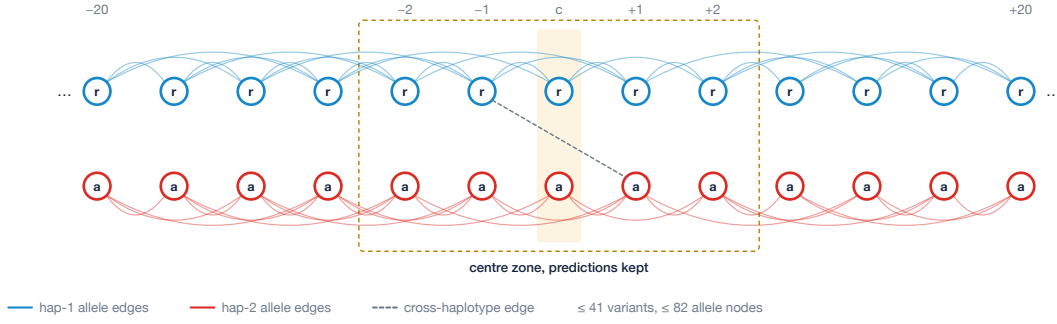


Supplementary Fig. 3: Read-based correction. **a**, Each read is assigned to a haplotype by majority over its phased SNV and SV alleles (indels count 0.1, 5mC 0); reads with majority ≤ 0.65 or fewer than two informative alleles remain untagged. **b**, Each variant's phase is re-derived from the tagged reads and retained only if the consistent fraction ρ exceeds 0.75; otherwise the variant is unphased. Only variants supported by both read populations keep a phase, which removes assignments driven by a few chimeric or mis-mapped reads. The two thresholds are `-readConfidence 0.65` and `-snpConfidence 0.75`.

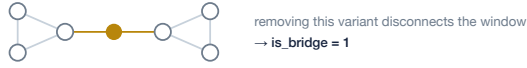


Supplementary Fig. 4: Overview of the graph neural network correction module. **a**, phase phases the called variants from the long-read alignments; every phased variant whose phasing entropy reaches 0.8 (amber) becomes the trigger of a window of ± 20 variants. **b**, The window is encoded as a graph whose nodes are the two alleles of each variant and whose edges are the surviving DOT edges (Supplementary Fig. 5d). Each allele node carries 31 features in four groups and each edge six features (Supplementary Tables 1 and 2); the shaded squares stand for the individual features of a group, not for their values. **c**, Each of four GPS layers passes the graph through a local branch, in which a node attends to its neighbours in the phasing graph (GATv2), and a global branch, in which every allele node in the window attends to every other. The two representations are fused additively with a skip connection from the layer input (dashed grey) and passed through a feed-forward block with a second skip connection. A classifier head then assigns each allele node a phase-error score, and the two allele nodes of a variant are averaged. **d**, Variants whose predicted phase-error score reaches 0.30 are unphased: the genotype is written unphased and without a phase set, the genotype itself is never flipped, and connectivity over the surviving edges splits the phase set in two. Detailed views of the window, the network and the phase-set update are given in Supplementary Figs. 5 and 6.

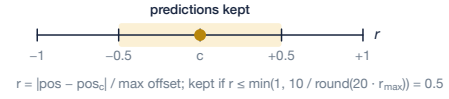
a Window around a trigger variant ($PE \geq 0.8$)



b Bridge-vertex feature



c Centre-zone mask

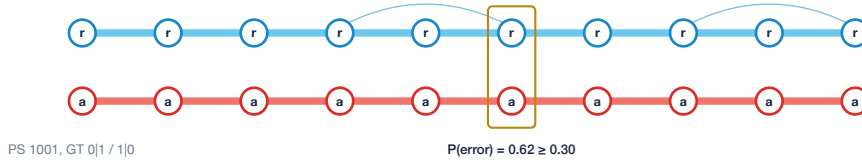


d Exported phasing graph (DOT)

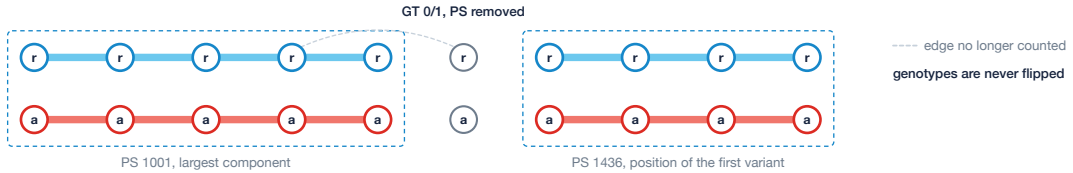
1001.1 → 1543.1 [label=16.0] 1001.2 → 1543.2 [label=1.1]

position.allele → position.allele, label = vote weight cast at the source

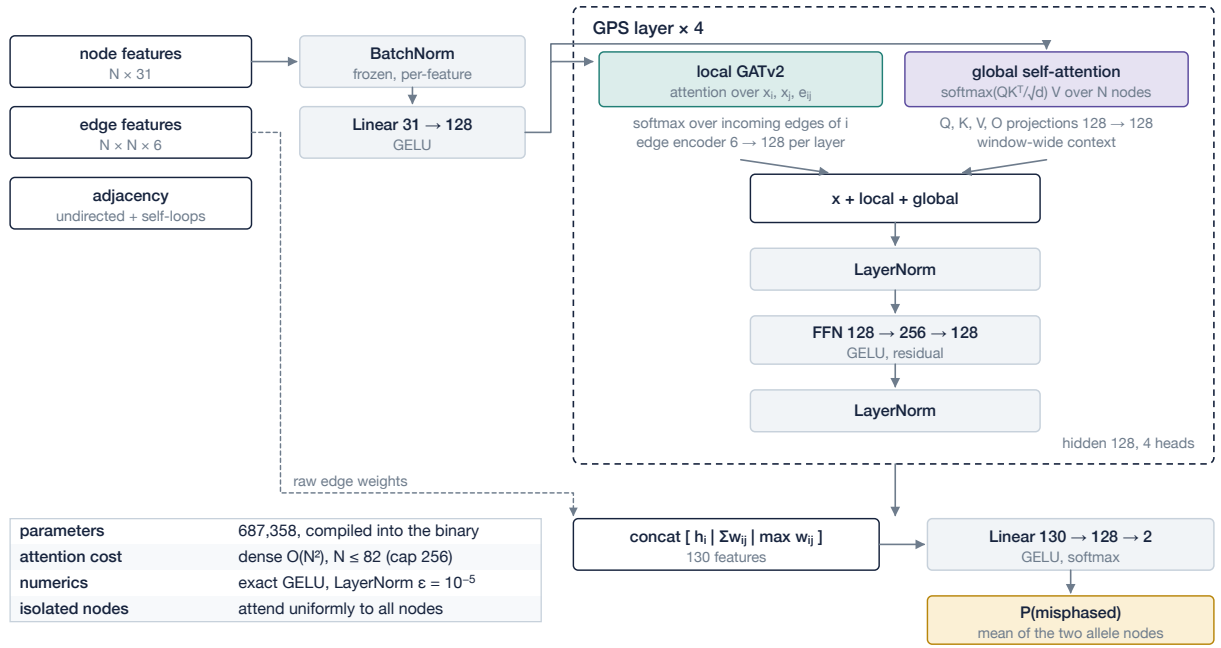
e Before correction: one phase set



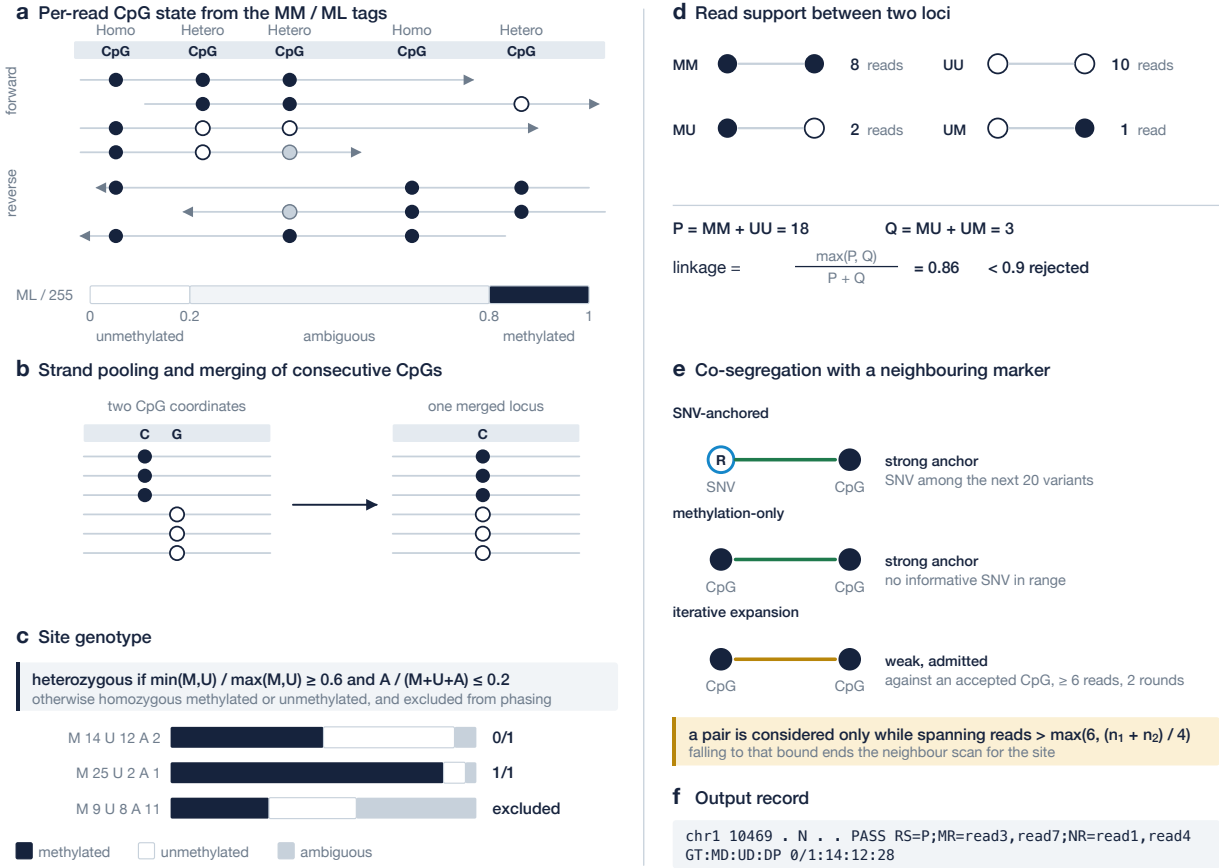
f After correction: variant unphased, phase set split



Supplementary Fig. 5: Window construction and phase-set update for the graph neural network. **a**, A window of ± 20 variants (two allele nodes each) is opened around every phased variant with $PE \geq 0.8$; edges are the DOT edges with both endpoints in the window (blue and red arcs) plus self-loops; a dashed grey edge marks a cross-haplotype link. Predictions are read only in the central zone. **b**, Bridge-vertex feature: a variant whose removal disconnects the window graph. **c**, Definition of the central zone. A window holds at most 41 variants and 82 allele nodes (a 256-node cap exists but is never reached at the default window size); SNV, indel, SV and 5mC records are merged and sorted by position before nodes are created. The retained zone is the central half of the window's genomic span: variants no further from the trigger than half the largest trigger distance in the window. The same rule, with the same constant, defines which labelled nodes carried a training loss, and the zone scored at inference is the zone the network was trained on. Edges are the DOT edges with both endpoints in the window, symmetrized, plus one self-loop per node whose features are the mean over that node's incident edges. The predictions of the two allele nodes of a variant are averaged, and a variant covered by several windows receives the mean over those windows. **d**, The phasing graph that the module reads, shown in its DOT export format (`-dot`): one line per voting pair, source and target written as position.allele, and the vote weight accumulated at the source as the label. **e**, Phase-set update, before correction: a variant with predicted phase-error score ≥ 0.30 (amber) inside a phase set. **f**, After correction the variant is written as an unphased genotype without PS; connectivity of the remaining variants is recomputed over the surviving DOT edges, the largest component keeps the original PS and every other component receives a new PS equal to the position of its first variant. Genotypes are never flipped.



Supplementary Fig. 6: Graph neural network architecture. Node features are standardized and projected to 128 dimensions. Each of four GPS layers combines a local GATv2 branch over the window graph (edge features enter through a per-layer edge encoder) with a global self-attention branch over all nodes, fused additively into the residual stream and followed by layer normalization and a feed-forward block. The final embedding is concatenated with the sum and the non-negative maximum, over the node's incoming edges excluding its self-loop, of the logistic-transformed edge weight $1/(1 + e^{-w})$, and classified into phased-correctly versus misphased; the probabilities of the two allele nodes of a variant are averaged.



Supplementary Fig. 7: Allele-specific methylation calling with modcall. **a**, Per-read CpG state is read from the MM/ML tags on both strands: methylated at $ML/255 \geq 0.8$, unmethylated at ≤ 0.2 , ambiguous in between. Only CpG dinucleotides are considered. **b**, 5mC is called on one strand at a time; reverse-strand calls are mapped to the forward CpG coordinate and runs of consecutive CpGs are merged into a single locus carrying one read list. This removes strand from the problem and lets **phase** vote on methylation loci with the unmodified graph algorithm. **c**, A site is heterozygous when $\min(M,U) / \max(M,U) \geq 0.6$ and $A / (M + U + A) \leq 0.2$, for methylated, unmethylated and ambiguous read counts M , U and A ; otherwise it is homozygous and excluded. **d**, Balance alone does not imply allele specificity, since stochastic methylation satisfies **c** as well. Read support between two loci is therefore decomposed into the four state pairs, whose concordant and discordant totals are exactly the quantities P and Q that the phasing graph accumulates; a pair is linked when $\max(P, Q) / (P + Q) \geq 0.9$. The example shown falls below the threshold and is rejected. **e**, Linking runs in two modes, against an SNV among the next 20 variants or, where no informative SNV is in range, against another candidate CpG, followed by an iterative expansion that admits weak candidates against an already accepted CpG. In every mode a pair is considered only while the reads spanning it exceed $\max(6, (n_1 + n_2) / 4)$, and falling to that bound ends the neighbour scan for that site rather than skipping one pair. **f**, Output record; the MR and NR read names attach the 5mC alleles to reads in **phase**.

a Switch errors, flips and Hamming distance within one block



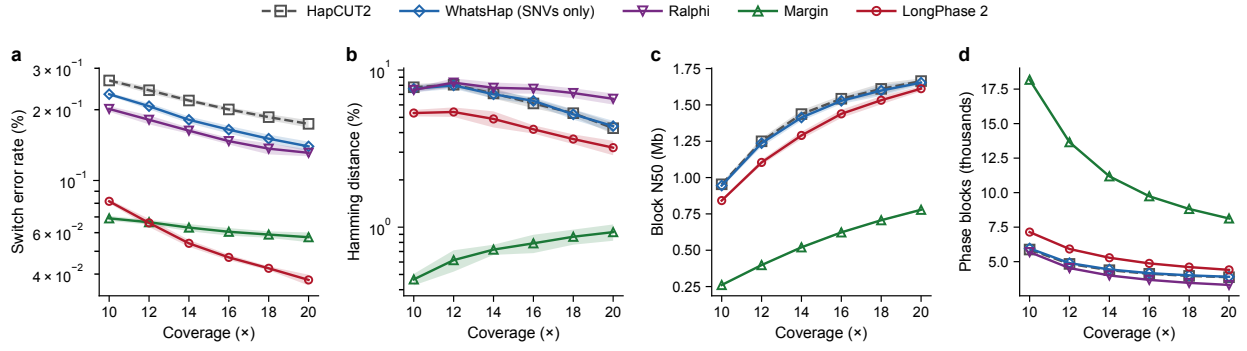
b Block statistics



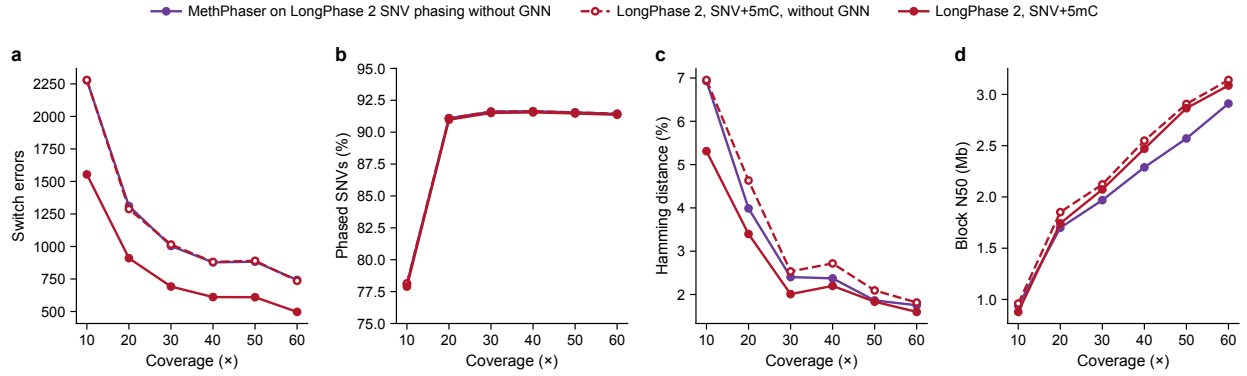
Supplementary Fig. 8: Phasing evaluation metrics. **a**, Within one intersected block, the query haplotype string is compared with the truth in switch encoding; the number of differing positions is the switch-error count, adjacent switch pairs are counted as flips, and the Hamming distance is the minimum over the two orientations of misassigned variants. **b**, Block intersection and the block-level statistics. Only variants that are heterozygous in both files and phased in both are compared (dark bars); blocks left with a single variant are ignored (outline). Switch error rate and Hamming distance are computed over the intersected blocks; the phased fraction counts truth heterozygous variants matched and phased in the query before the partition, and the block N50 is computed on the query's own blocks of at least two variants, as implemented in `longphase compare`.

Placeholder: panels to be regenerated with the submitted version.

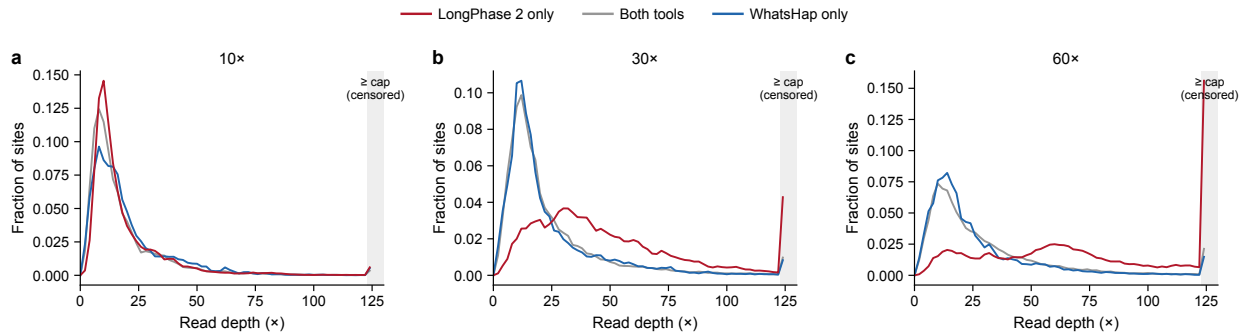
Supplementary Fig. 9: Calibration of the voting and read-filtering constants. Each panel is a sweep on HG002 at 10–60 $\times$; the value adopted in LongPhase 2 is marked. **a**, Distribution of pair ambiguity $s = \min(P, Q) / \max(P, Q)$ over all assessed pairs and over the subset that participate in a switch error, showing that both mass and residual error concentrate below $s = 0.1$ and motivating $\alpha = 0.1$ as the boundary of the up-weighted class. **b**, Switch errors and block N50 as a function of the no-vote threshold β . Because switch errors fall monotonically as β decreases, switch error alone does not identify an optimum; $\beta = 0.7$ is the smallest value at which block N50 is not yet degraded, and is adopted on that basis. **c**, Switch errors as a function of the up-weight w applied when $s \leq \alpha$, at each coverage; the reduction saturates by $w \approx 20$. **d**, Distribution of base quality over all bases and over bases carrying a sequencing error, showing errors concentrated below Q10 and motivating $\theta = 12$; and switch errors as a function of θ . **e**, Switch errors over the grid of single-read-guard parameters N (number of single-read votes that triggers the guard) and μ (ambiguity cut applied to the surviving votes), and switch errors with and without the guard at each coverage. [Placeholder: produce or remove. The five panels exist in the group's thesis material (Po-Hui Tzeng, 2026, Figs. 3.7, 3.10–3.12, 3.14, 3.17, 3.18, 3.21) and would be regenerated with the submitted version on the PEPPER-Margin-DeepVariant calls and coverages used here, in one consistent style; state two caveats: in **b** the switch-error curve has no interior optimum and β rests on the N50 trade-off, and in **e** the guard slightly *increased* switch errors at 10 $\times$. If the figure is removed, delete the sentences citing it in Methods and Supplementary Method 3.]



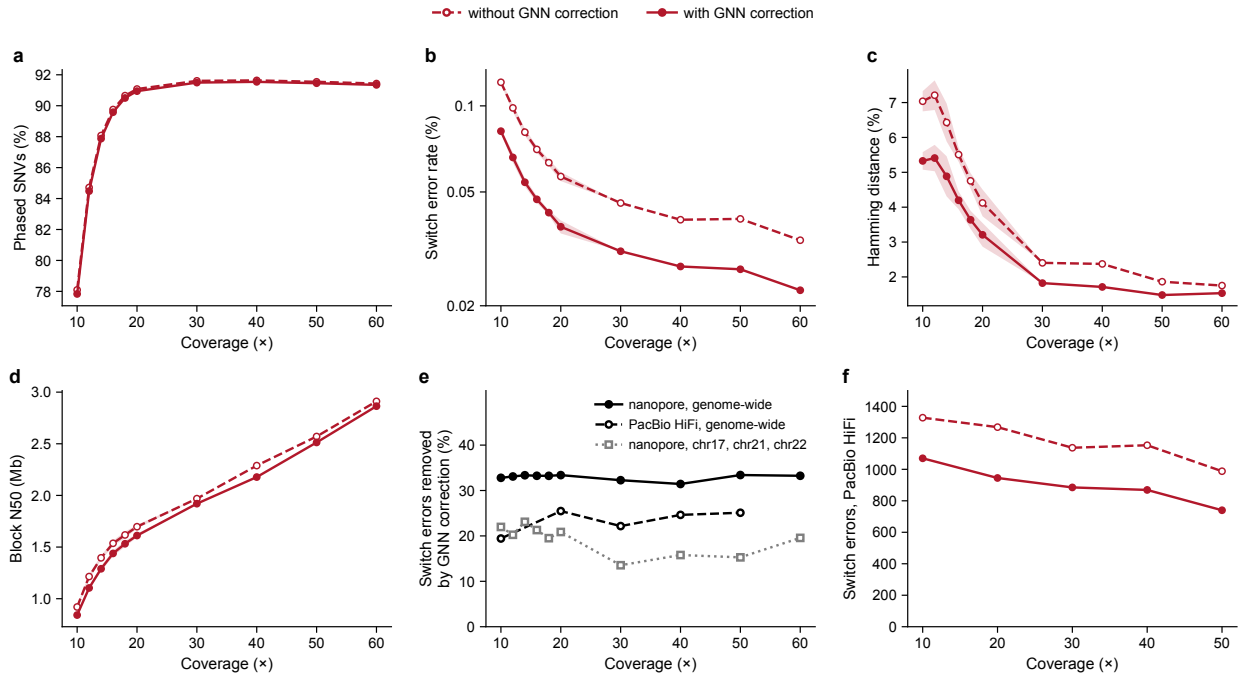
Supplementary Fig. 10: SNV-only phasing compared with Margin and Ralphi. HG002 nanopore R10.4.1 data at 10–20 $\times$, means of ten replicates with s.d., scored against v5.0q; LongPhase 2 with GNN correction. **a**, Switch error rate. **b**, Hamming distance. **c**, Block N50; Ralphi is omitted because its blocks sum to more than the autosome length. **d**, Number of phase blocks. Values in Supplementary Table 8.



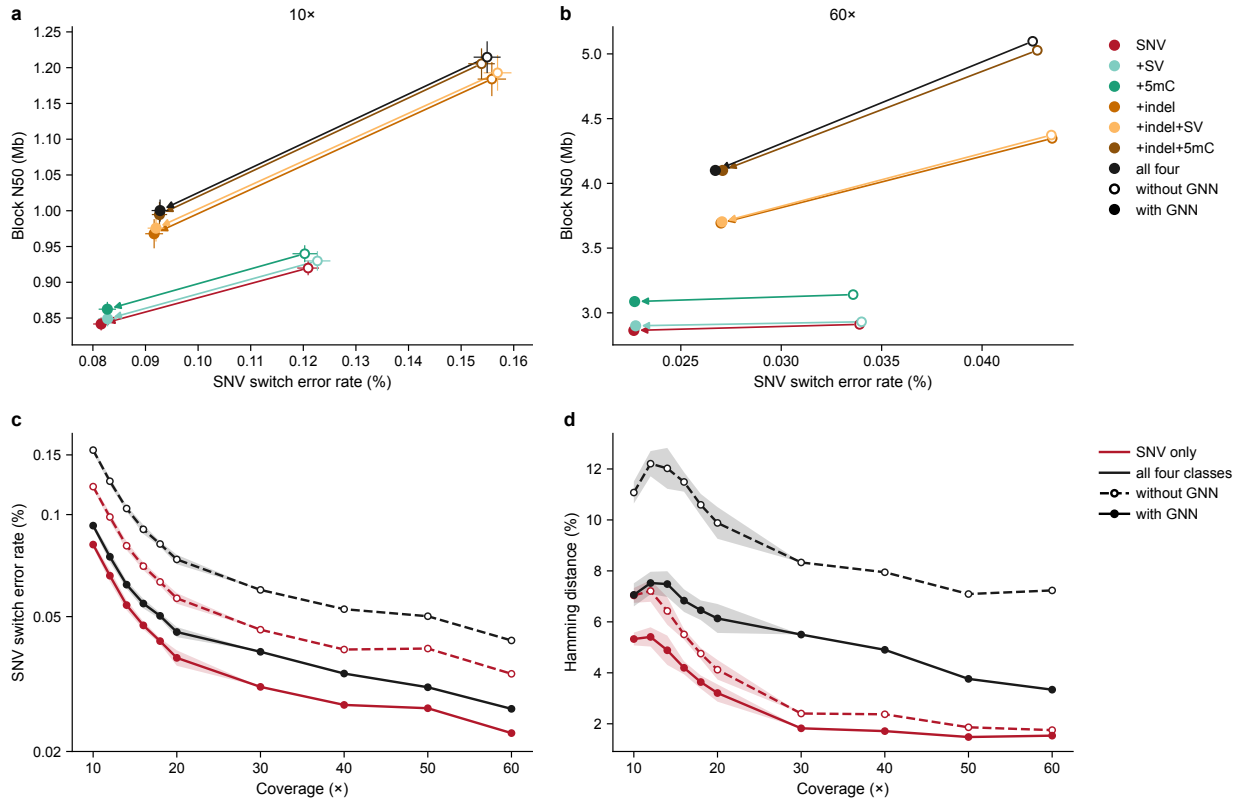
Supplementary Fig. 11: Joint methylation co-phasing compared with MethPhaser. HG002 nanopore R10.4.1 data, replicate 1, scored on SNVs against v5.0q. MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction. **a**, Switch errors. **b**, Fraction of benchmark heterozygous SNVs phased. **c**, Hamming distance. **d**, Block N50.



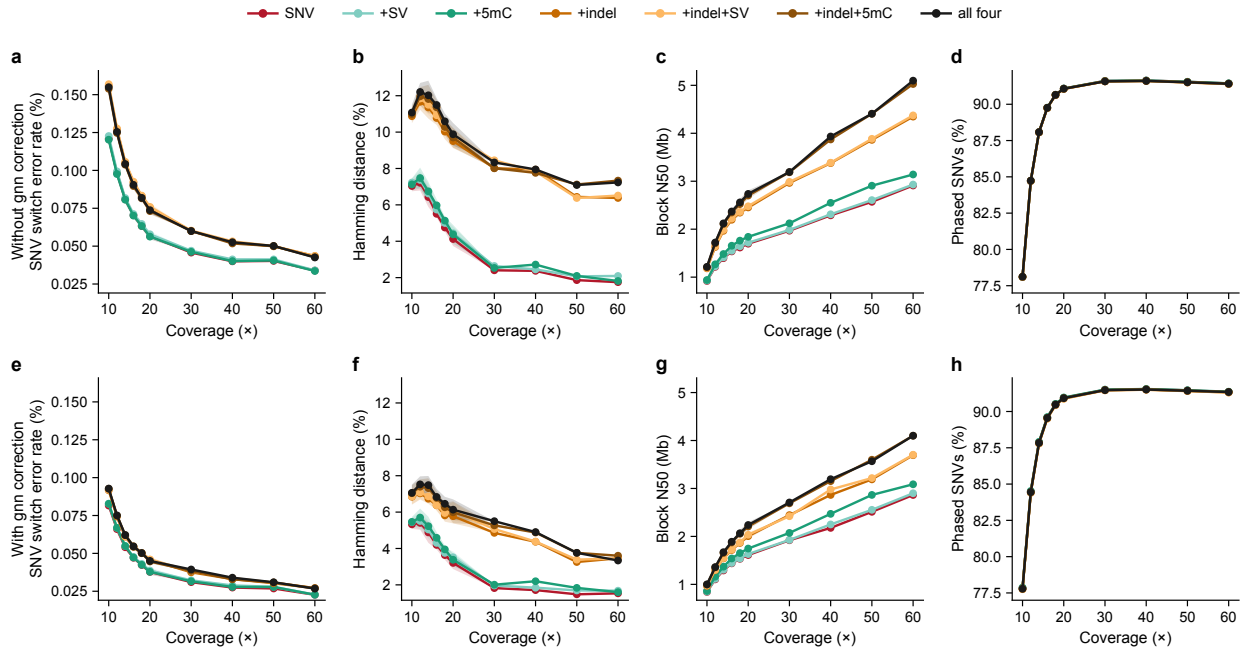
Supplementary Fig. 12: Read depth at heterozygous SNV calls left unphased by LongPhase 2, by WhatsHap or by both. HG002 nanopore R10.4.1 data, replicate 1, at 10 $\times$ (a), 30 $\times$ (b) and 60 $\times$ (c); 2 $\times$ bins. The grey bin collects every site at or above the depth cap of the variant caller (about 125 $\times$), whose true depth is censored. Sets as in Fig. 6a.



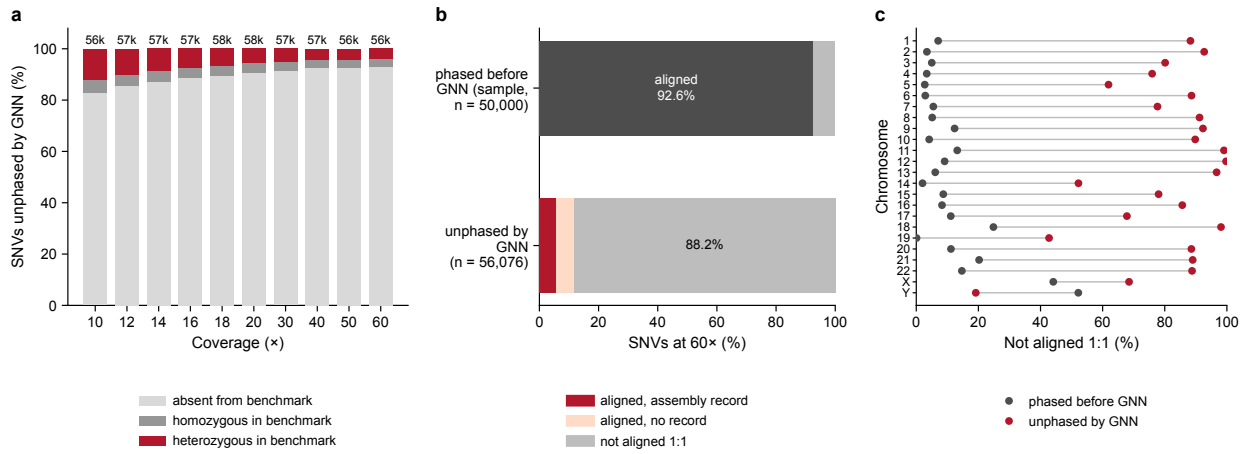
Supplementary Fig. 13: Effect of GNN correction on SNV-only phasing. LongPhase 2 with and without GNN correction; HG002 nanopore R10.4.1 data unless stated; means of ten replicates with s.d. at 10–20×; one down-sampling at 30–60×. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate. **c**, Hamming distance. **d**, Block N50. **e**, Fraction of switch errors removed by GNN correction on nanopore and PacBio HiFi data, genome-wide, and on the nanopore chromosomes held out from GNN training (chr17, chr21 and chr22; Supplementary Table 15). **f**, Switch errors on PacBio HiFi data, with the network trained on nanopore data.



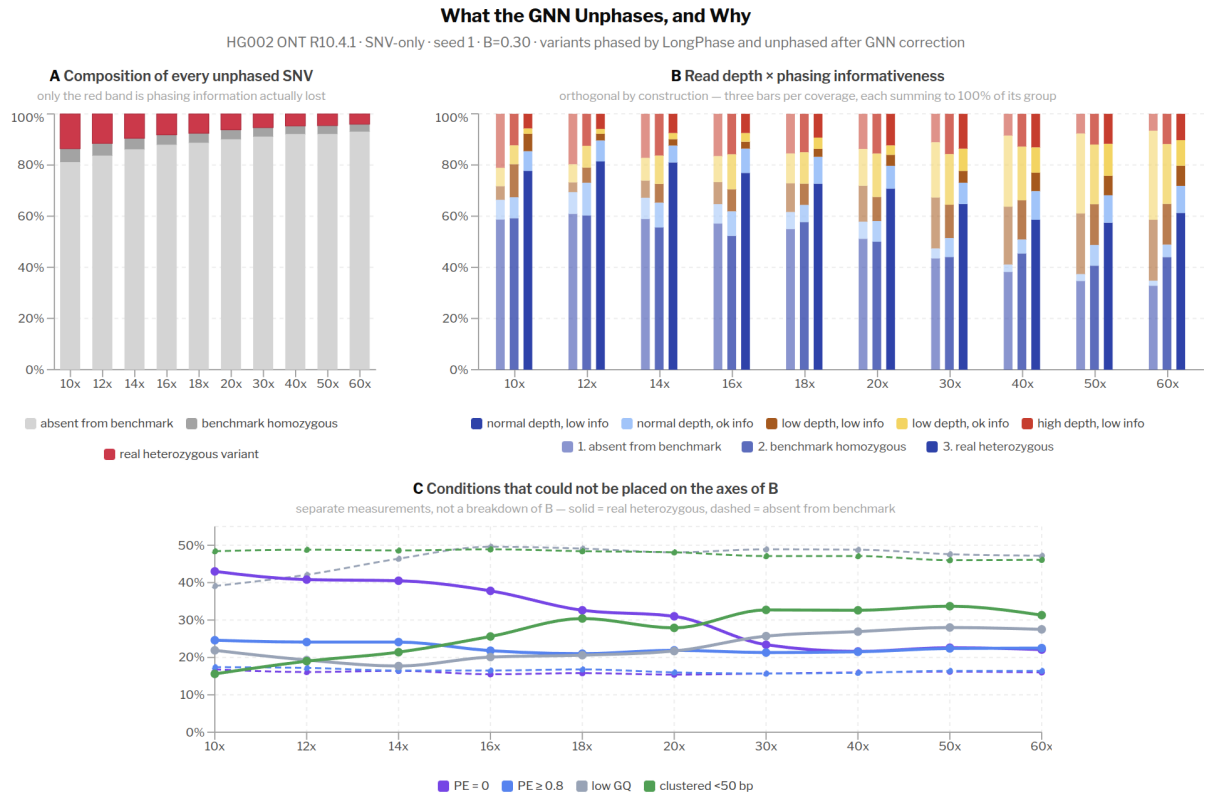
Supplementary Fig. 14: Co-phasing configurations with and without GNN correction. HG002 nanopore R10.4.1 data; seven combinations of variant classes, each phased separately; SNV metrics against v5.0q. **a,b**, SNV switch error rate against block N50 at 10× (**a**; means of ten replicates with s.d.) and 60× (**b**); arrows run from the result without (open) to the result with (filled) GNN correction. **c,d**, SNV switch error rate (**c**) and Hamming distance (**d**; computed over SNVs and indels for four-class phasing) for SNV-only and four-class phasing.



Supplementary Fig. 15: All co-phasing configurations across coverage. HG002 nanopore R10.4.1 data; means of ten replicates with s.d. at 10–20×; one down-sampling at 30–60×. Top row without, bottom row with GNN correction. **a,e**, SNV switch error rate. **b,f**, Hamming distance (over SNVs and indels where indels are phased). **c,g**, Block N50. **d,h**, Fraction of benchmark heterozygous SNVs phased. Values in Supplementary Table 10.



Supplementary Fig. 16: SNVs unphased by GNN correction. HG002 nanopore R10.4.1 data, SNV-only phasing, replicate 1, final GNN model. **a**, Classification against the v5.0q benchmark VCF, matched by position without a benchmark BED; numbers above bars, total unphased. **b**, Status in the T2T-HG002 v1.1 assembly of the SNVs unphased at 60× and of a random sample of SNVs phased before GNN correction: inside or outside the regions where both assembled haplotypes align 1:1 to GRCh38, and, inside, whether an assembly variant record starts at the same coordinate. **c**, Per chromosome, fraction of each set outside the 1:1 alignment. Values in Supplementary Table 12. [Rebuild from one site-level table (autosomes only, control excluding removed sites, allele-aware assembly comparison) before submission.]



The GNN unphases roughly 60,000 SNVs at every coverage, but 86–96% of them are calls the benchmark does not support as heterozygous, so almost none of the apparent loss is real phasing information (A); at 60× only 4.0% (2,411 variants) are genuine heterozygous sites. Placing each variant on two orthogonal axes (B) shows that true heterozygotes are removed overwhelmingly at normal read depth with weak phasing informativeness (61–81%), whereas benchmark-absent calls shift towards low depth as coverage rises (12.5% at 10× to 58.6% at 60×), the signature of hard-to-map regions. No variant is unphased for high depth alone: the high-depth / normal-informativeness cell is empty at every coverage. Panel C reports conditions that overlap too heavily to sit on those axes, as plain prevalences within each group. Among the true heterozygotes the GNN removes, 43.0% at 10× have a PE of exactly zero — against 16.8% of benchmark-absent calls — so no fixed entropy threshold could have found them; that share falls to 22.1% by 60× as phasing evidence improves. Clustering and low genotype quality run the other way, staying near 46–49% among benchmark-absent calls at every coverage while remaining well below that among true heterozygotes, which is what marks them as caller artefacts rather than phasing failures.

Supplementary Fig. 17: Read depth, phasing informativeness and phasing entropy of the SNVs unphased by GNN correction. SNV-only phasing, seed 1. Panel a is superseded by Supplementary Fig. 16a. **b**, Within the benchmark-absent, homozygous and heterozygous categories, distribution over read depth and phasing informativeness. **c**, Prevalence of phasing entropy of exactly 0, phasing entropy ≥ 0.8 , low genotype quality and clustering within 50 bp; solid, benchmark heterozygous sites; dashed, benchmark-absent calls. [Raster carried over from the former main Fig. 6 (pptx), with uppercase panel letters and a caption inside the artwork; regenerate panels b and c from their source table, define the depth and informativeness bins (Methods), and drop panel a and the embedded caption.]

References

- [1] Rampášek, L. *et al.* Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*, vol. 35 (2022).
- [2] Veličković, P. *et al.* Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).
- [3] Brody, S., Alon, U. & Yahav, E. How attentive are graph attention networks? In *International Conference on Learning Representations (ICLR)* (2022).
- [4] Vaswani, A. *et al.* Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).
- [5] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)* (2017).
- [6] Patterson, M. *et al.* WhatsHap: weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology* **22**, 498–509 (2015).
- [7] Shafin, K. *et al.* Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nature Methods* **18**, 1322–1332 (2021).
- [8] Hansen, N. F. *et al.* A complete diploid human genome benchmark for personalized genomics. *Cell* **189**, 4857–4875.e31 (2026).
- [9] Olson, N. D. *et al.* The Genome in a Bottle HG002 assembly-based variant benchmark set enables comprehensive benchmarking of small and structural variants. *bioRxiv* (2026).
- [10] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).
- [11] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).
- [12] Zheng, Z. *et al.* Clair3-RNA: a deep learning-based small variant caller for long-read RNA sequencing data. *Nature Communications* **16** (2025).
- [13] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).
- [14] Kolesnikov, A. *et al.* Local read haplotagging enables accurate long-read small variant calling. *Nature Communications* **15** (2024).
- [15] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).
- [16] Oxford Nanopore Technologies. epi2me-labs/wf-human-variation: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. <https://github.com/epi2me-labs/wf-human-variation> (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).
- [17] Pacific Biosciences. HiFi-somatic-WDL: tumor-normal variant calling workflow using HiFi reads. <https://github.com/PacificBiosciences/HiFi-Somatic-WDL> (2025).
- [18] Cao, S. *et al.* cuteHap: haplotype-aware structural variant detection in phased long-read sequencing data. *Advanced Science* **13** (2026).
- [19] Wang, S. *et al.* A scalable framework for comprehensive typing of polymorphic immune genes from long-read data. *Advanced Science* **13** (2026).

- [20] Gabbutt, C. *et al.* Fluctuating DNA methylation tracks cancer evolution at clinical scale. *Nature* **645**, 764–773 (2025).
- [21] O’Neill, K. *et al.* Long-read sequencing of an advanced cancer cohort resolves rearrangements, unravels haplotypes, and reveals methylation landscapes. *Cell Genomics* **4**, 100674 (2024).
- [22] LaFlamme, C. W. *et al.* Diagnostic utility of DNA methylation analysis in genetically unsolved pediatric epilepsies and CHD2 epismutation refinement. *Nature Communications* **15** (2024).
- [23] Iwasaki, Y. *et al.* Bidirectional disruption of *GNAS* transcripts causes broad methylation defects in pseudohypoparathyroidism type 1B. *Proceedings of the National Academy of Sciences* **122** (2025).
- [24] Zalusky, M. P. *et al.* 3-hour genome sequencing and targeted analysis to rapidly assess genetic risk. *Genetics in Medicine Open* **2**, 101833 (2024).
- [25] Smits, D. J. *et al.* Nanopore long-read sequencing for the critically ill facilitates ultrarapid diagnostics and urgent clinical decision making. *European Journal of Human Genetics* **34**, 108–118 (2025).
- [26] Fleming, A. *et al.* Combined approaches, including long-read sequencing, address the diagnostic challenge of HYDIN in primary ciliary dyskinesia. *European Journal of Human Genetics* **32**, 1074–1085 (2024).
- [27] Thomas, R. *et al.* Splice-modulating antisense oligonucleotides targeting a pathogenic intronic variant in adult polyglucosan body disease correct mis-splicing and restore enzyme activity in patient cells. *Nucleic Acids Research* **53** (2025).
- [28] Tian, Y., McDonnell, S. K., Wu, L., Larson, N. B. & Wang, L. Fine mapping regulatory variants by characterizing native CpG methylation with nanopore long-read sequencing. *Human Genetics and Genomics Advances* **7**, 100532 (2026).
- [29] Galvez, J. H. *et al.* Benchmarking of Oxford Nanopore whole genome sequencing for germline variant and CpG methylation detection across Canada’s national platform for genome sequencing and analysis. *Frontiers in Genetics* **17** (2026).
- [30] Flack, N. *et al.* Chromosome-level, nanopore-only genome and allele-specific DNA methylation of Pallas’s cat, *Otocolobus manul*. *NAR Genomics and Bioinformatics* **5** (2023).
- [31] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).
- [32] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).
- [33] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).
- [34] Zhou, Y., Leung, A. W.-S., Ahmed, S. S., Lam, T.-W. & Luo, R. Duet: SNP-assisted structural variant calling and phasing using Oxford nanopore sequencing. *BMC Bioinformatics* **23** (2022).